FAKULTA APLIKOVANÝCH VĚD
ZÁPADOČESKÉ UNIVERZITY
V PLZNI

Edgebreaker

Komprese trojúhelníkových sítí

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1. Motivace

1.1 Dobový kontext

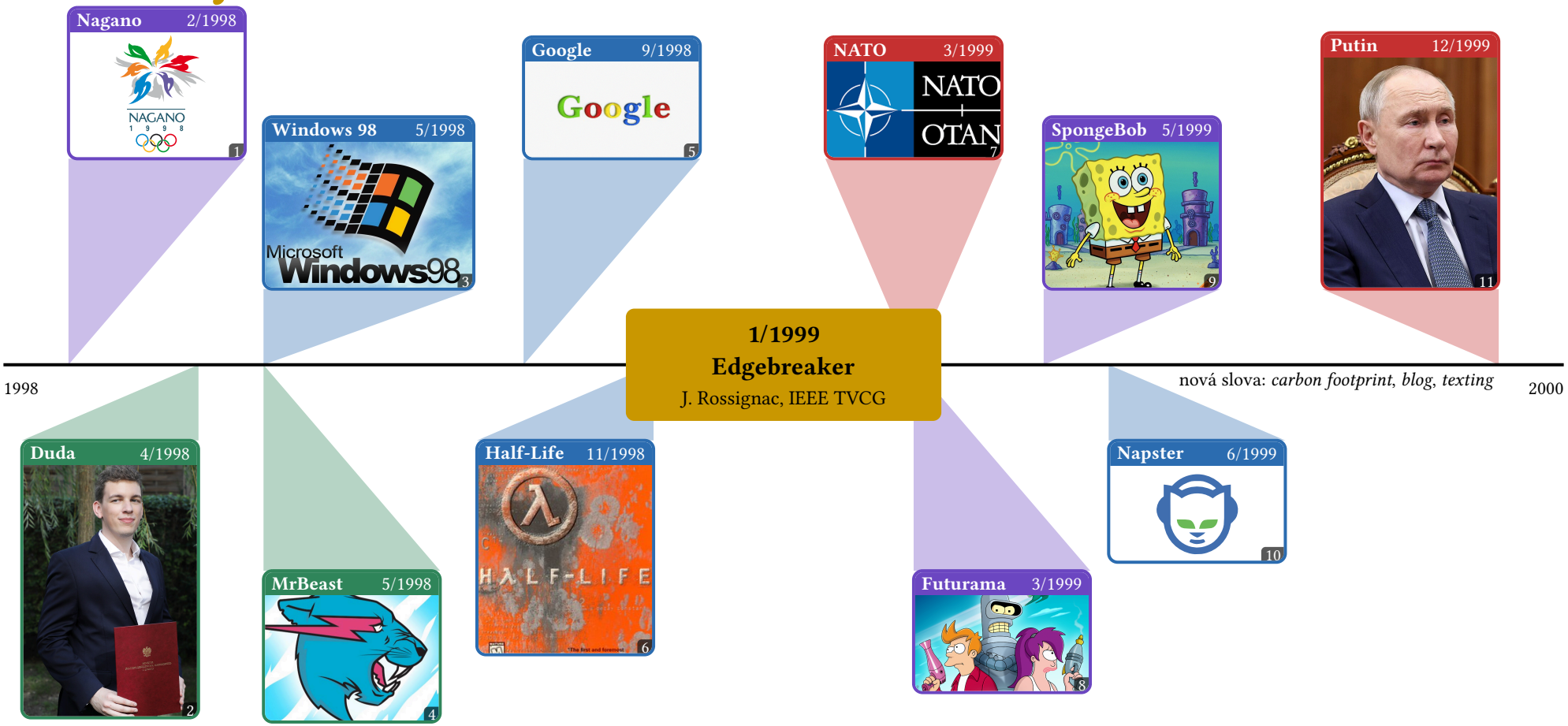


Figure 1: Dobový kontext vzniku algoritmu Edgebreaker.

1.1 Dobový kontext



Figure 2: Jarek Rossignac, zdroj: http://kelvinkuo.com/wp-content/uploads/2010/03/KKK_6258.jpg

1.2 Dělení

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z)$$

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.2.2 Konektivita

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.2.2 Konektivita

$$\forall e \in E : e = \left(e_i, e_j\right)$$

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.2.2 Konektivita

$$\forall e \in E : e = \left(e_i, e_j\right)$$

$$\forall t \in T : t = \left(v_i, v_j, v_k\right)$$

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.2.2 Konektivita

$$\forall e \in E : e = \left(e_i, e_j\right)$$

$$\forall t \in T : t = \left(v_i, v_j, v_k\right)$$

$$\forall c \in 3T : c = \left(c_{\mathrm{op}}\right)$$

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.2.2 Konektivita

$$V : 2F : 3E$$

2. Komprese

2.1 Pravidla

History =

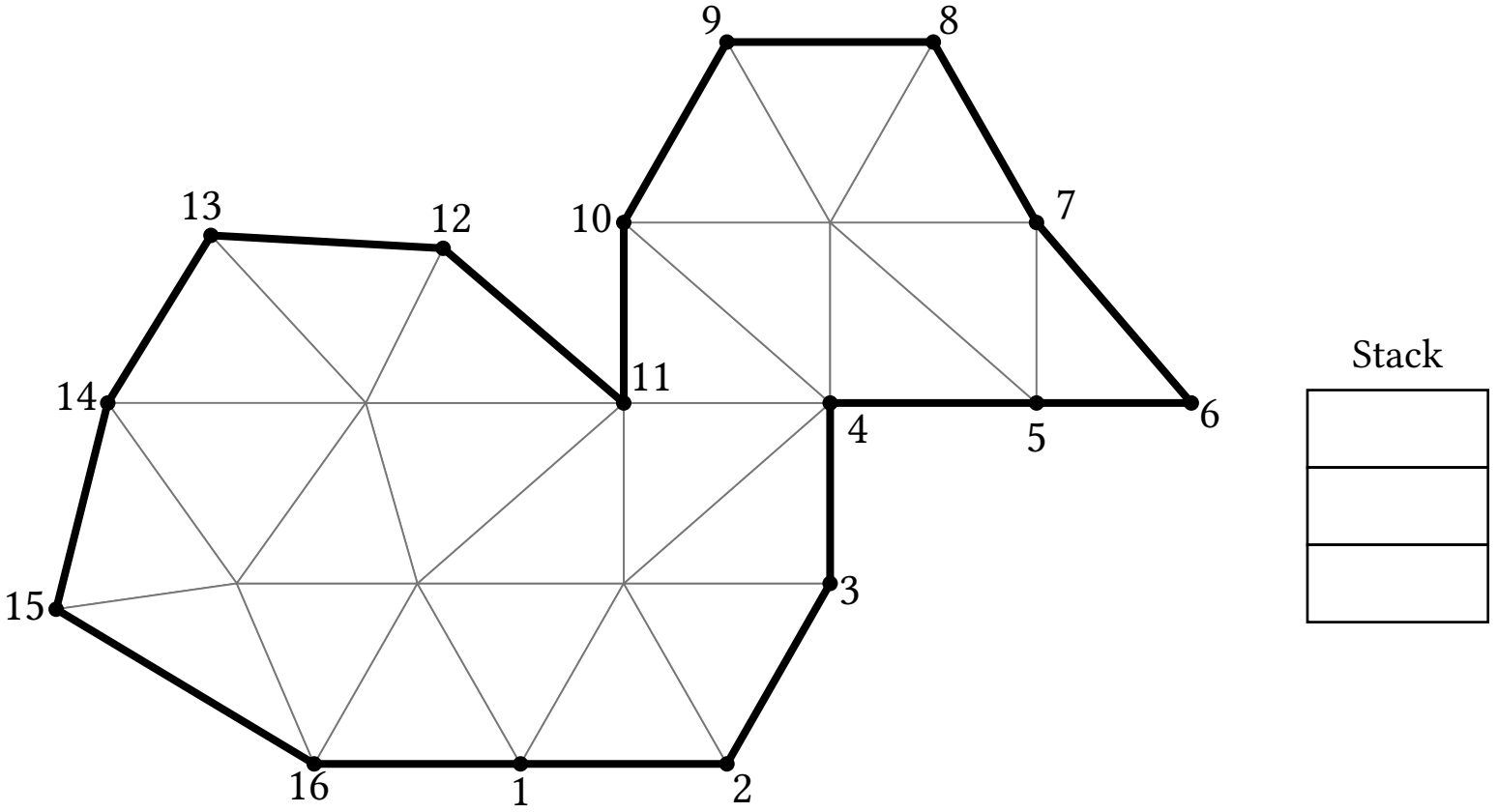


Figure 3: Jednoduchá komprese.

History = C

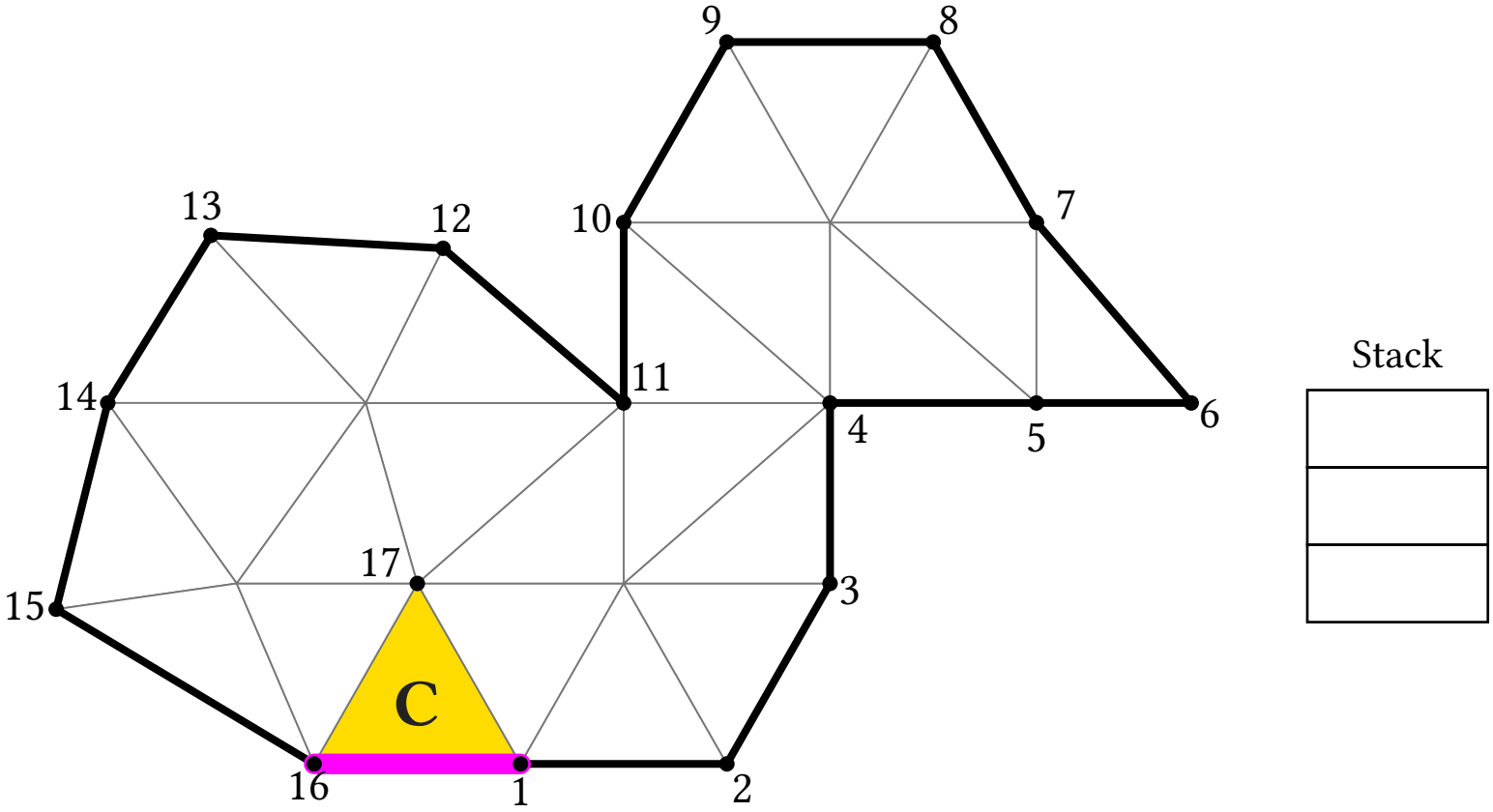


Figure 3: Jednoduchá komprese.

History = CC

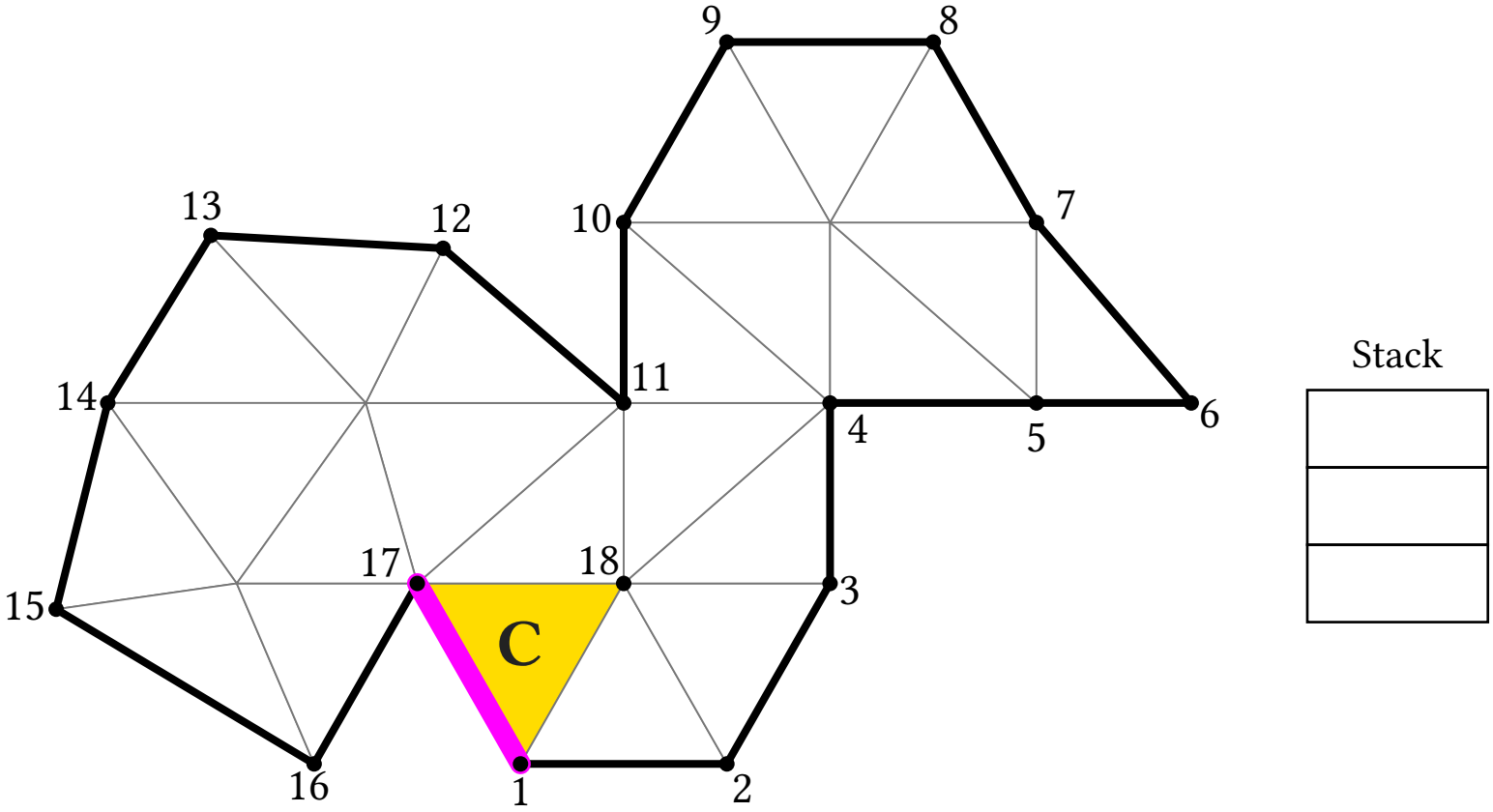


Figure 3: Jednoduchá komprese.

History = CCR

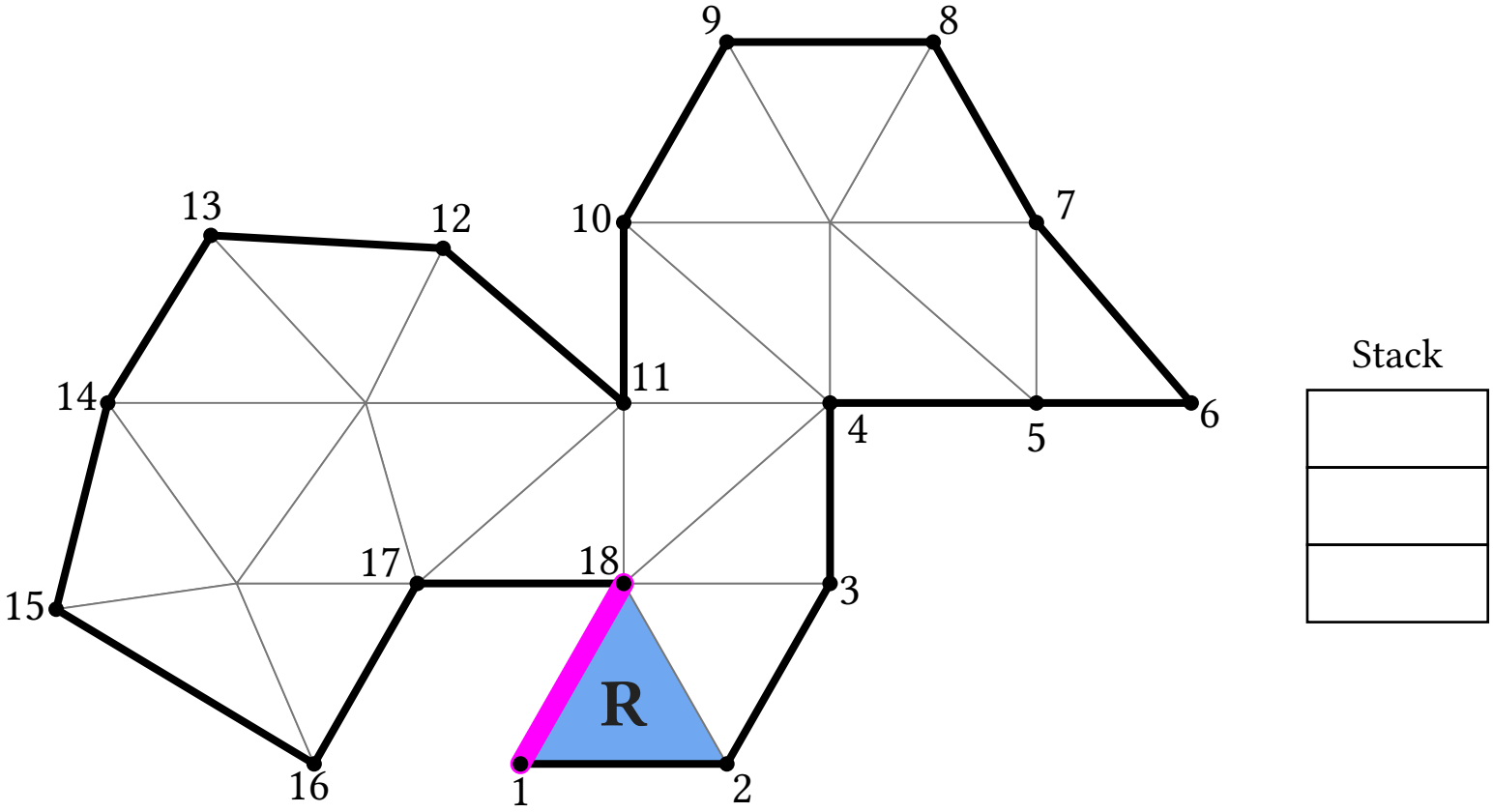


Figure 3: Jednoduchá komprese.

History = CCRR

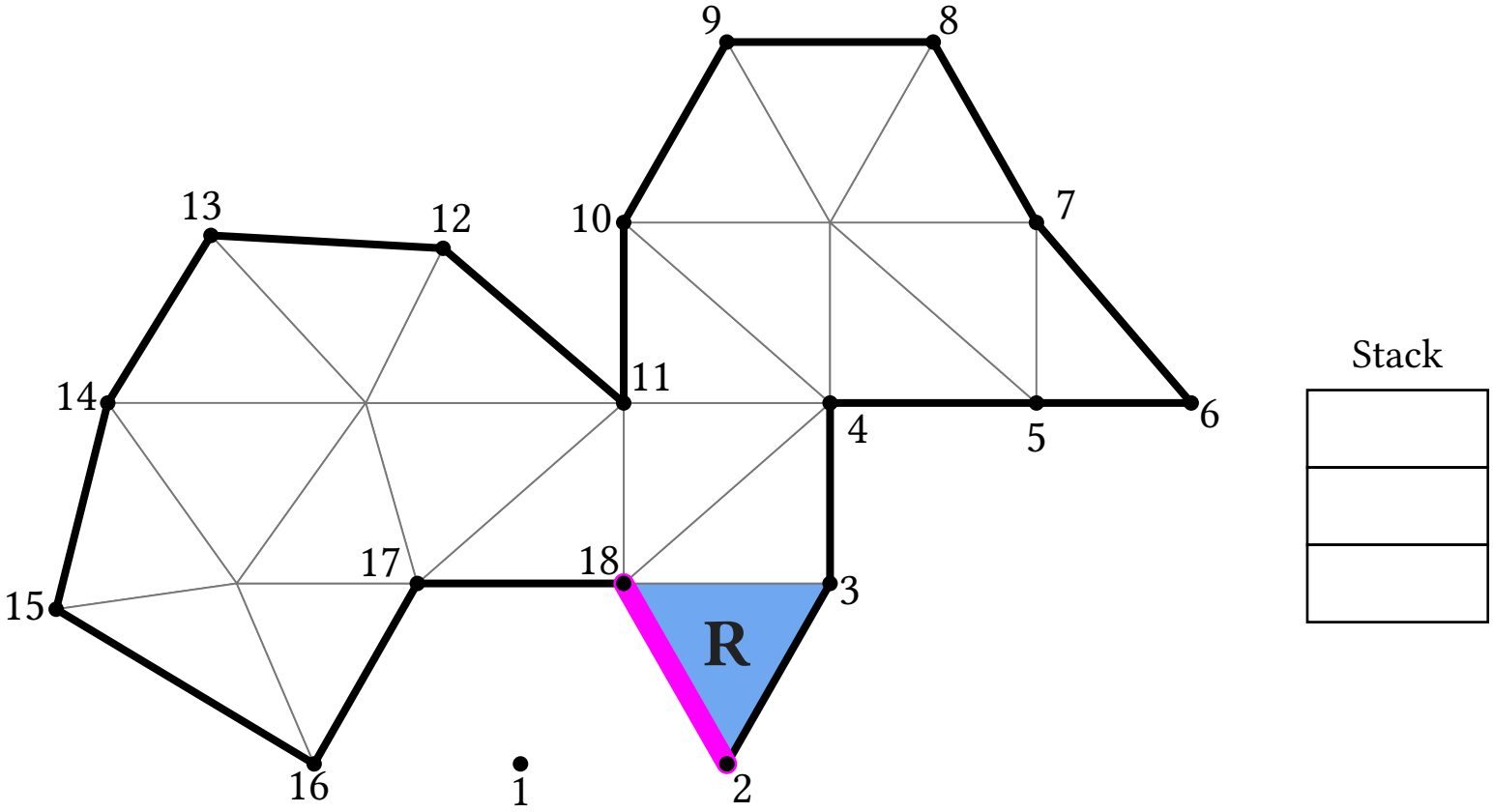


Figure 3: Jednoduchá komprese.

History = CCRRR

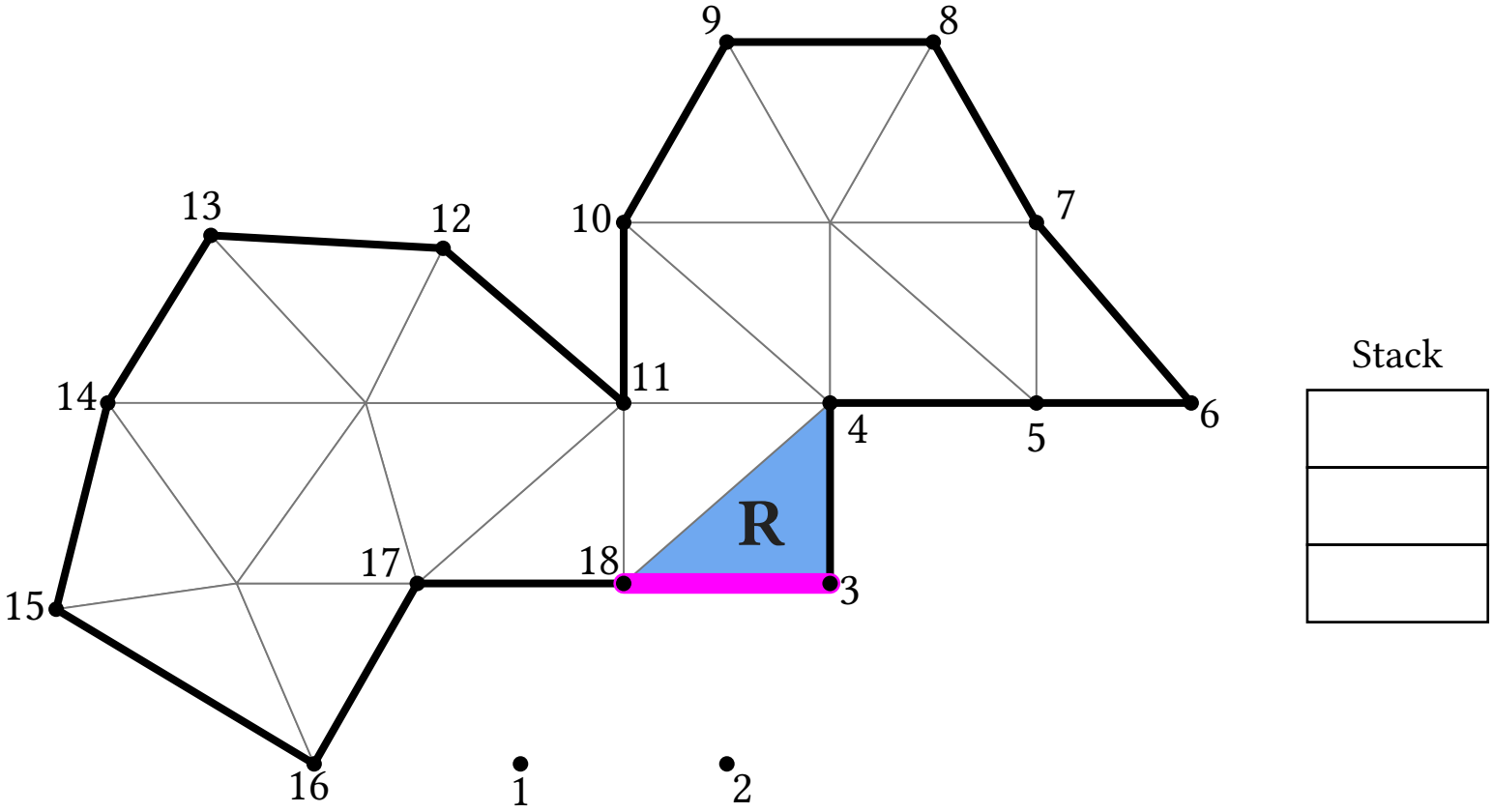


Figure 3: Jednoduchá komprese.

History = CCRRRS

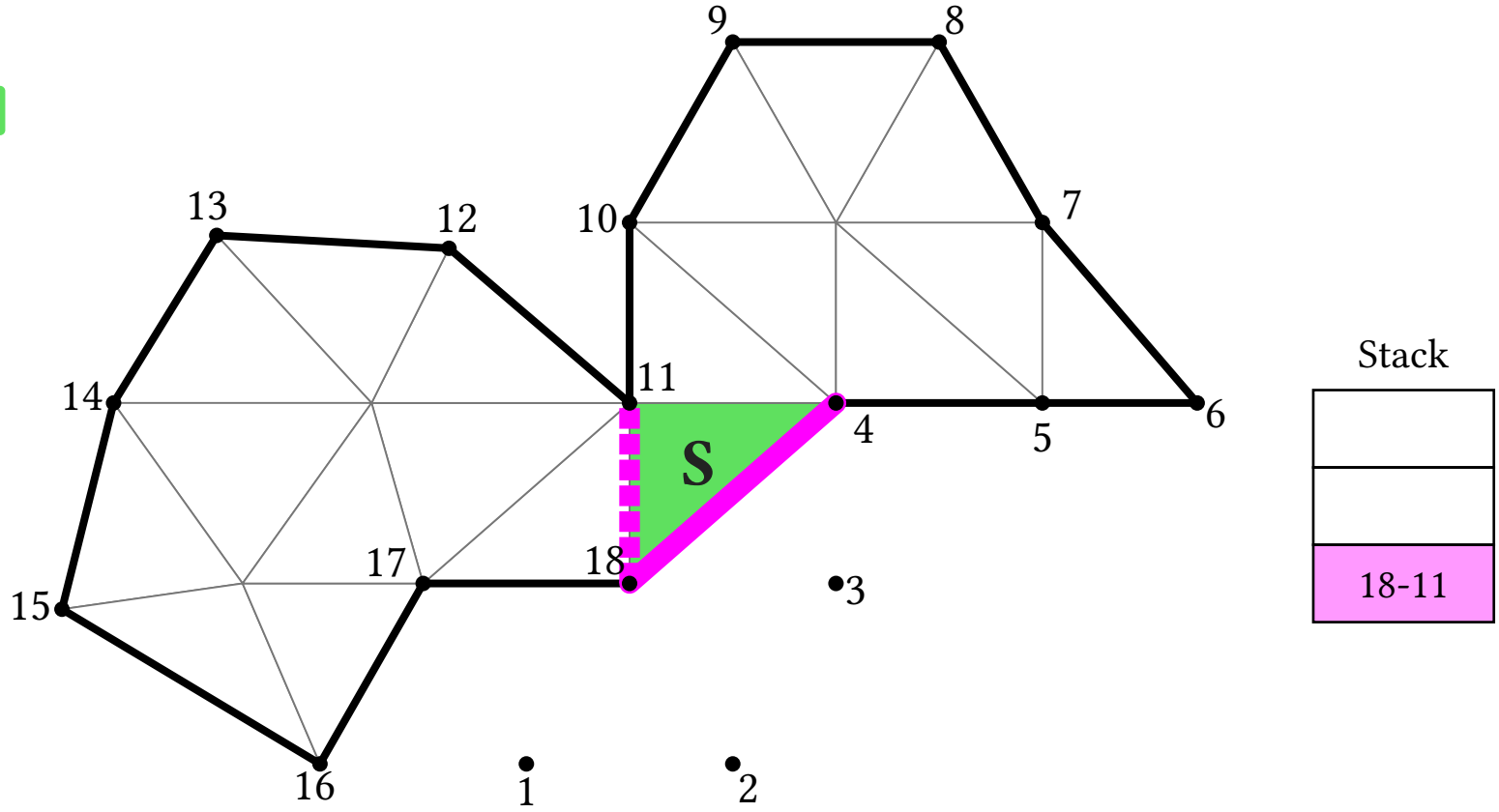


Figure 3: Jednoduchá komprese.

History = CCRRRSL

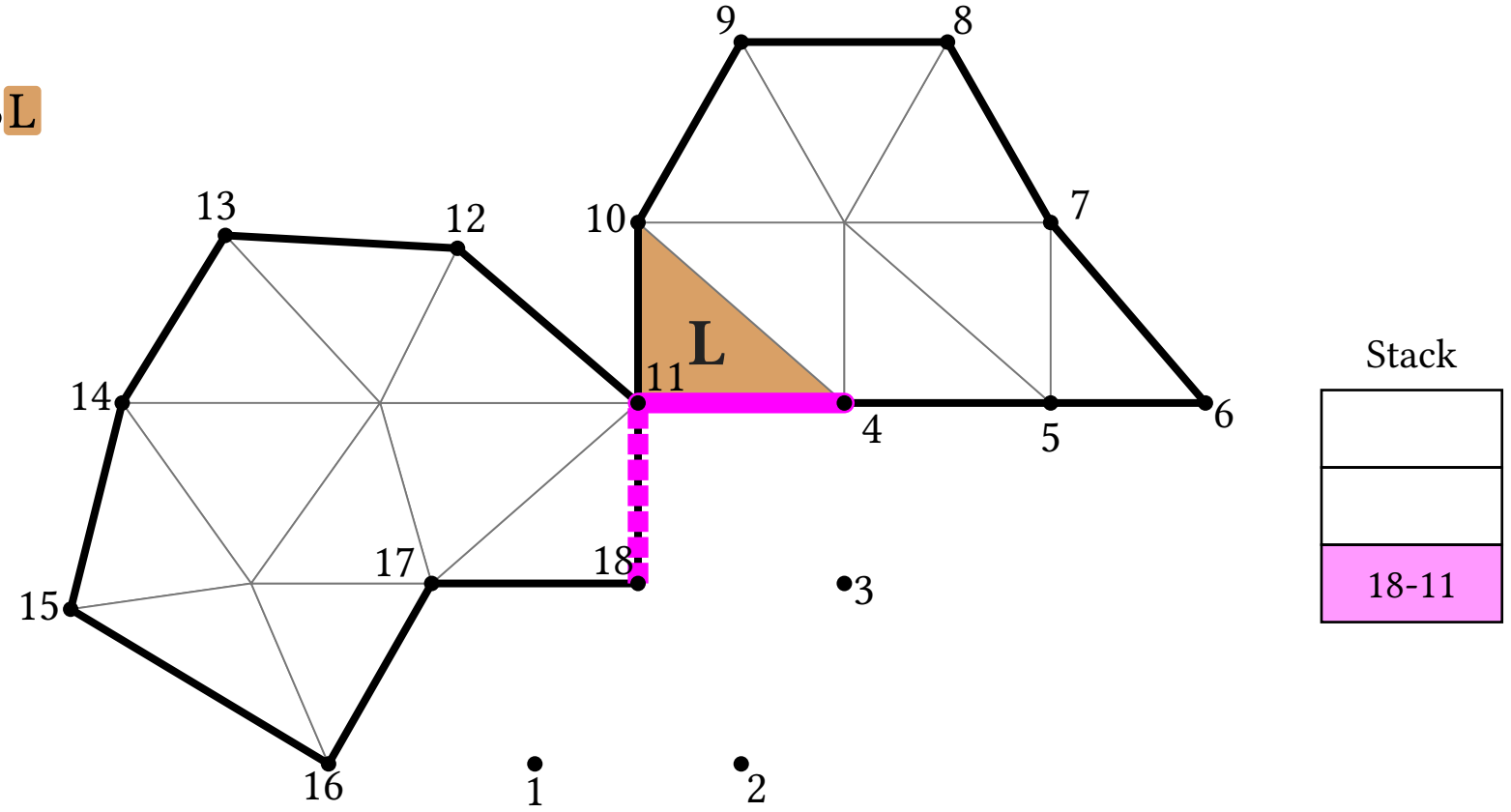


Figure 3: Jednoduchá komprese.

History = CCRRRSLC

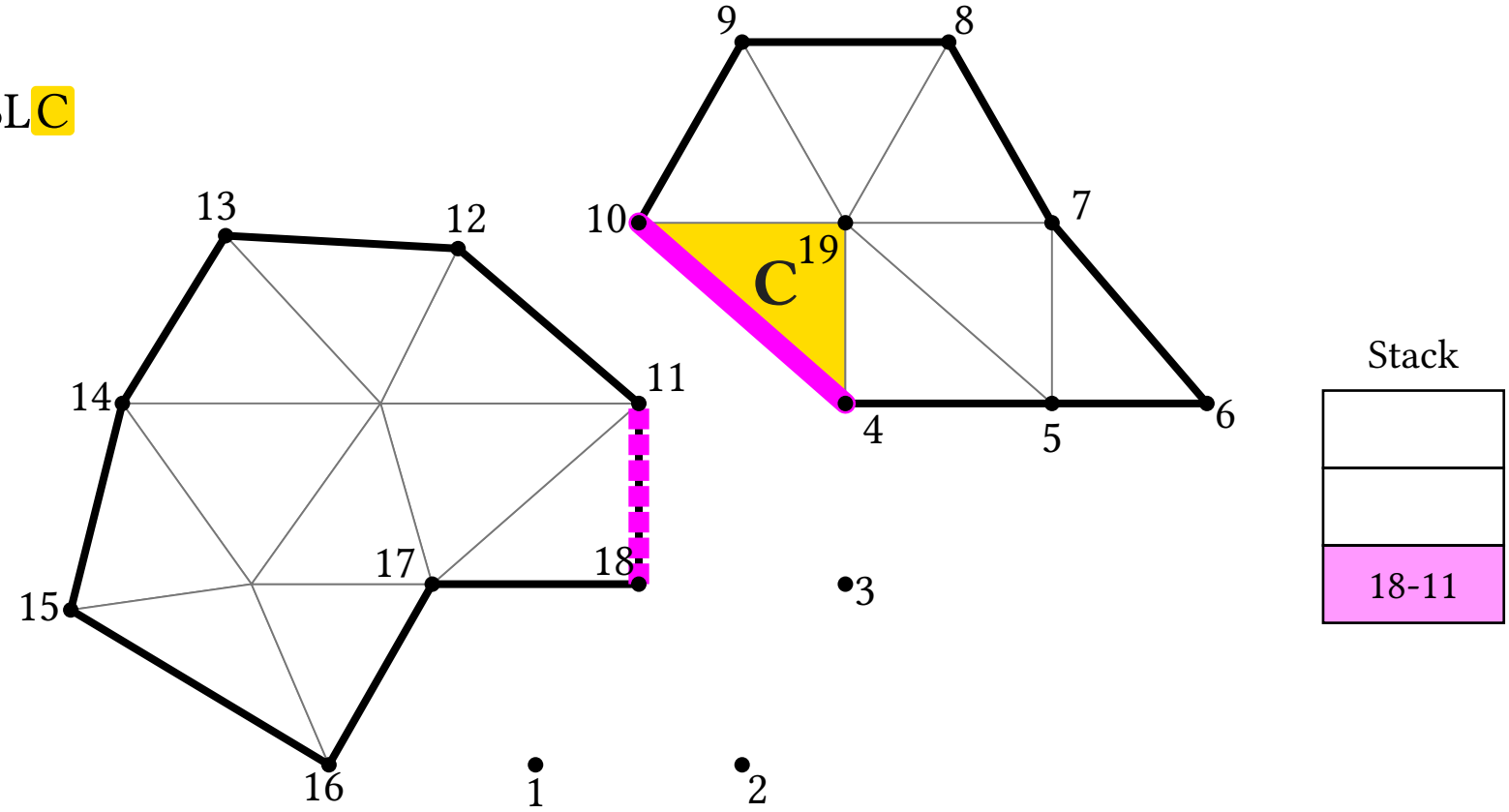


Figure 3: Jednoduchá komprese.

History = CCRRRSLCR

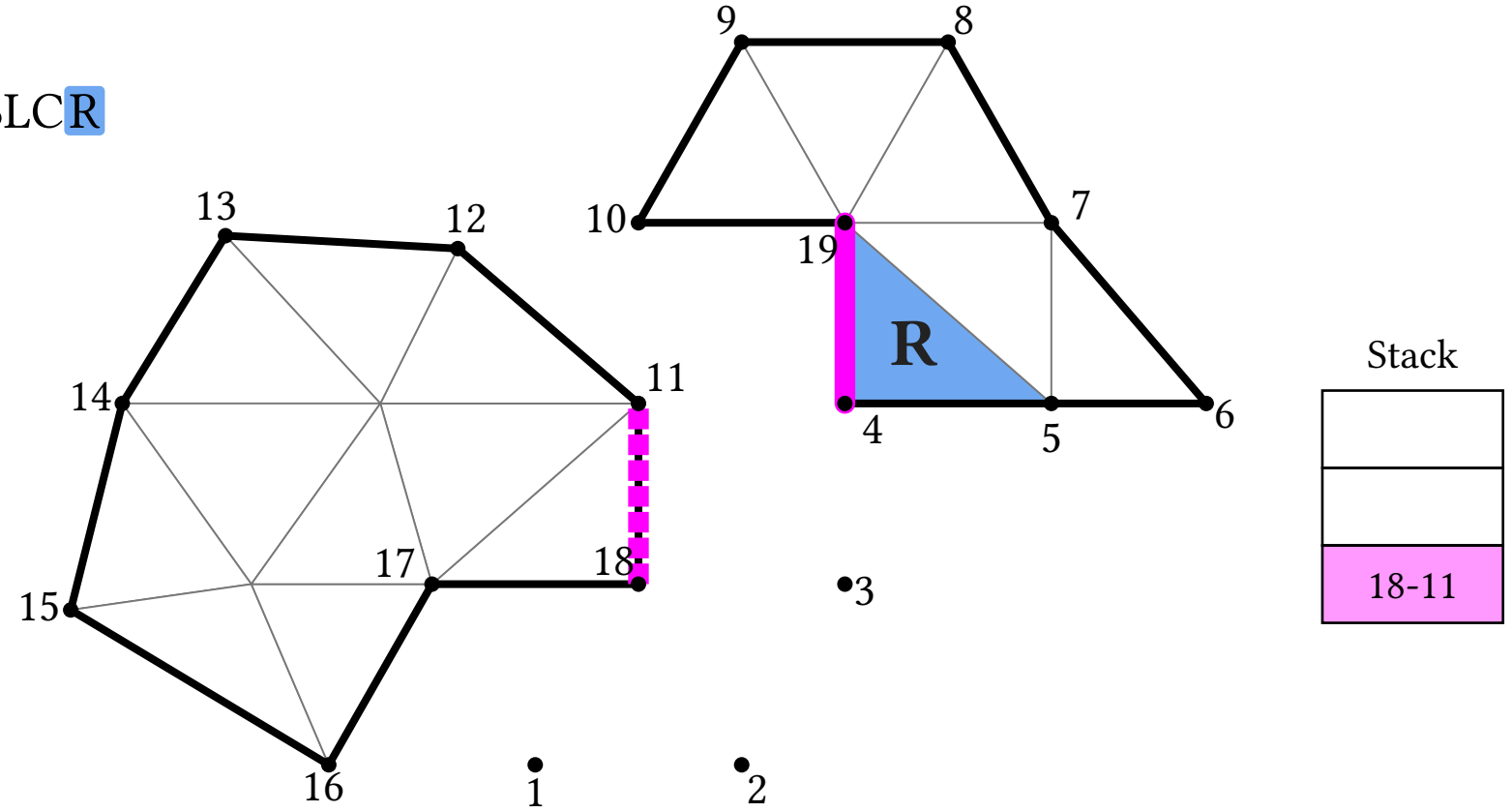


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRS

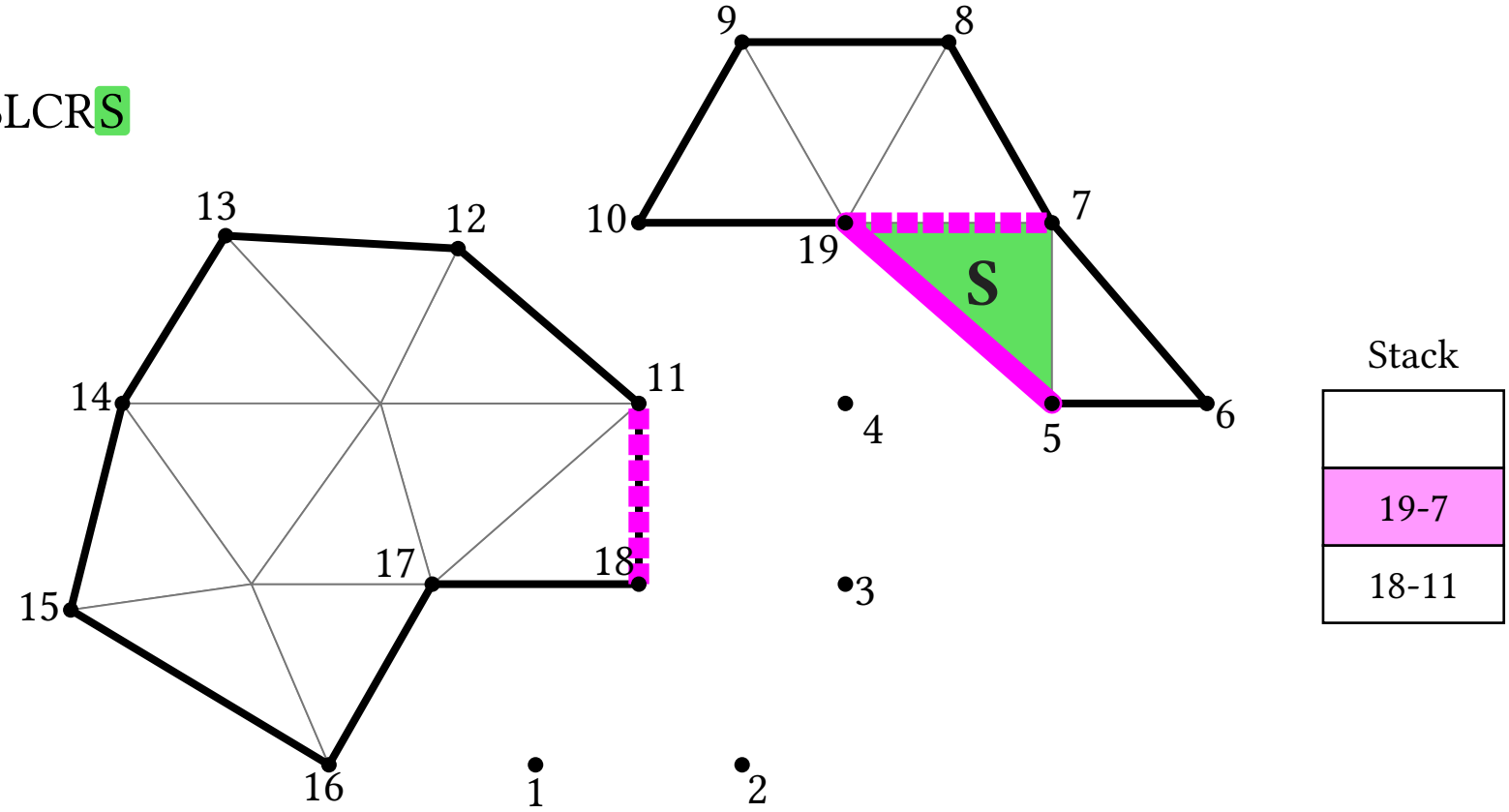


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSE

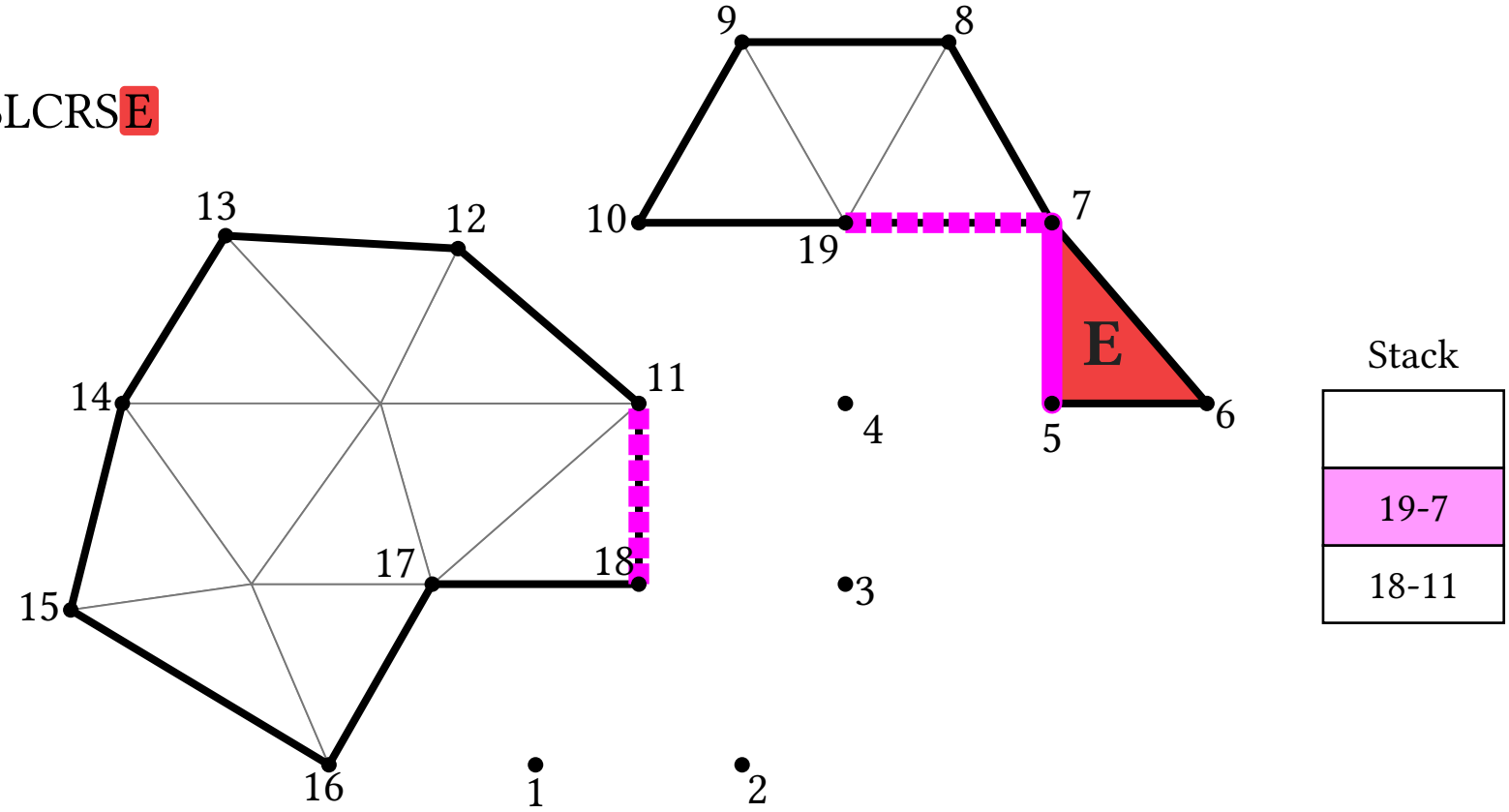


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSER

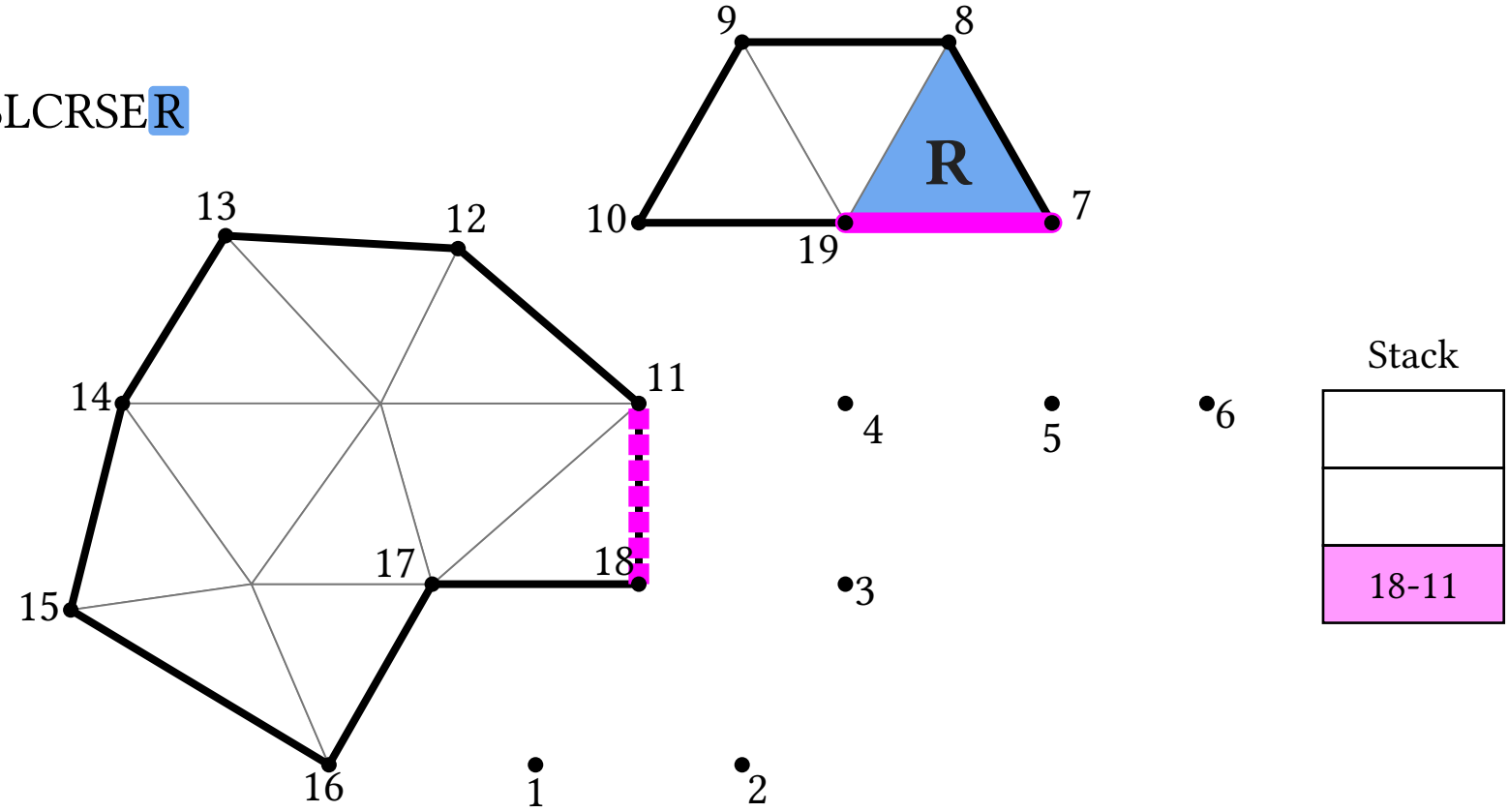


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERR

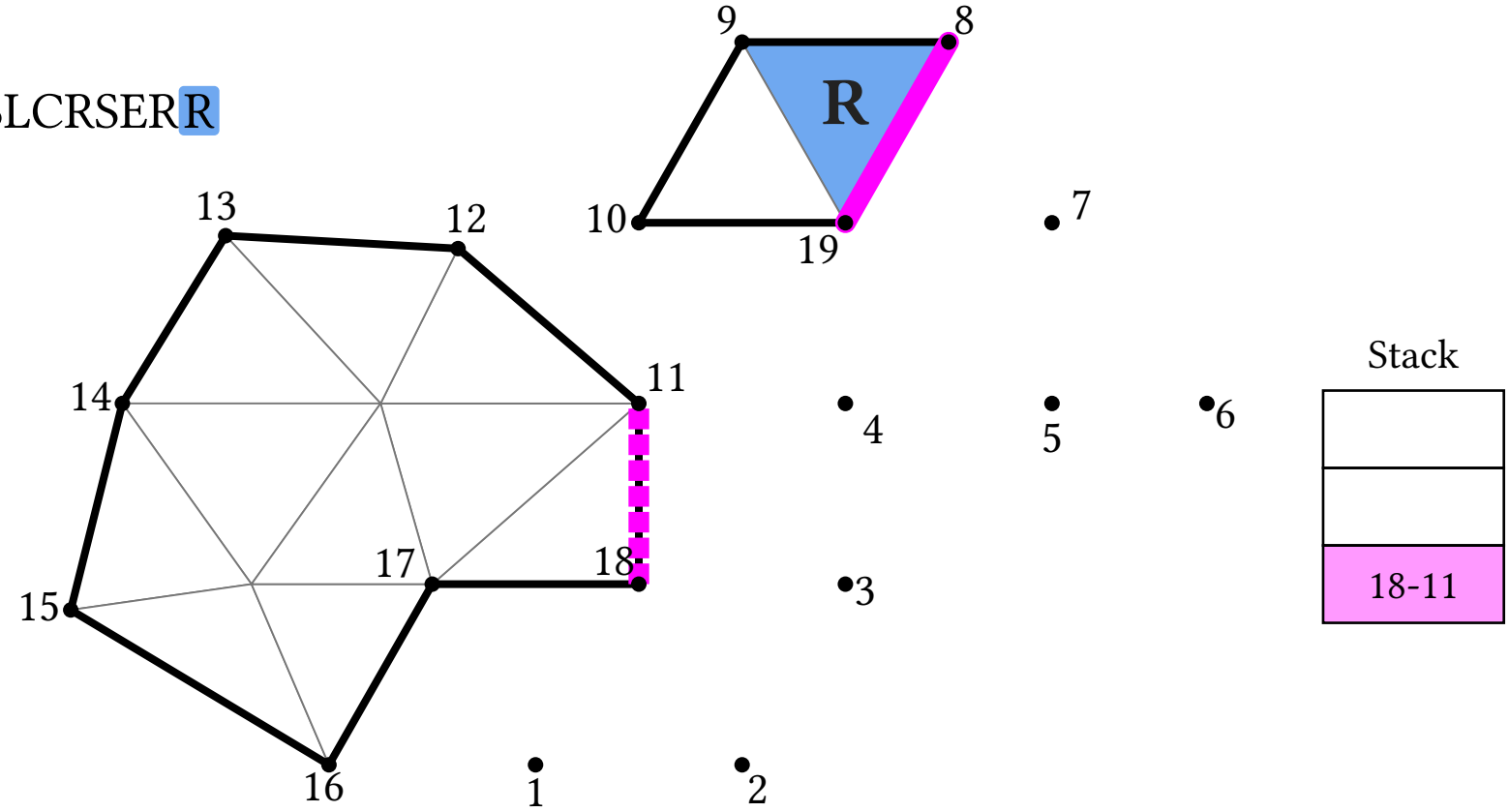


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRE

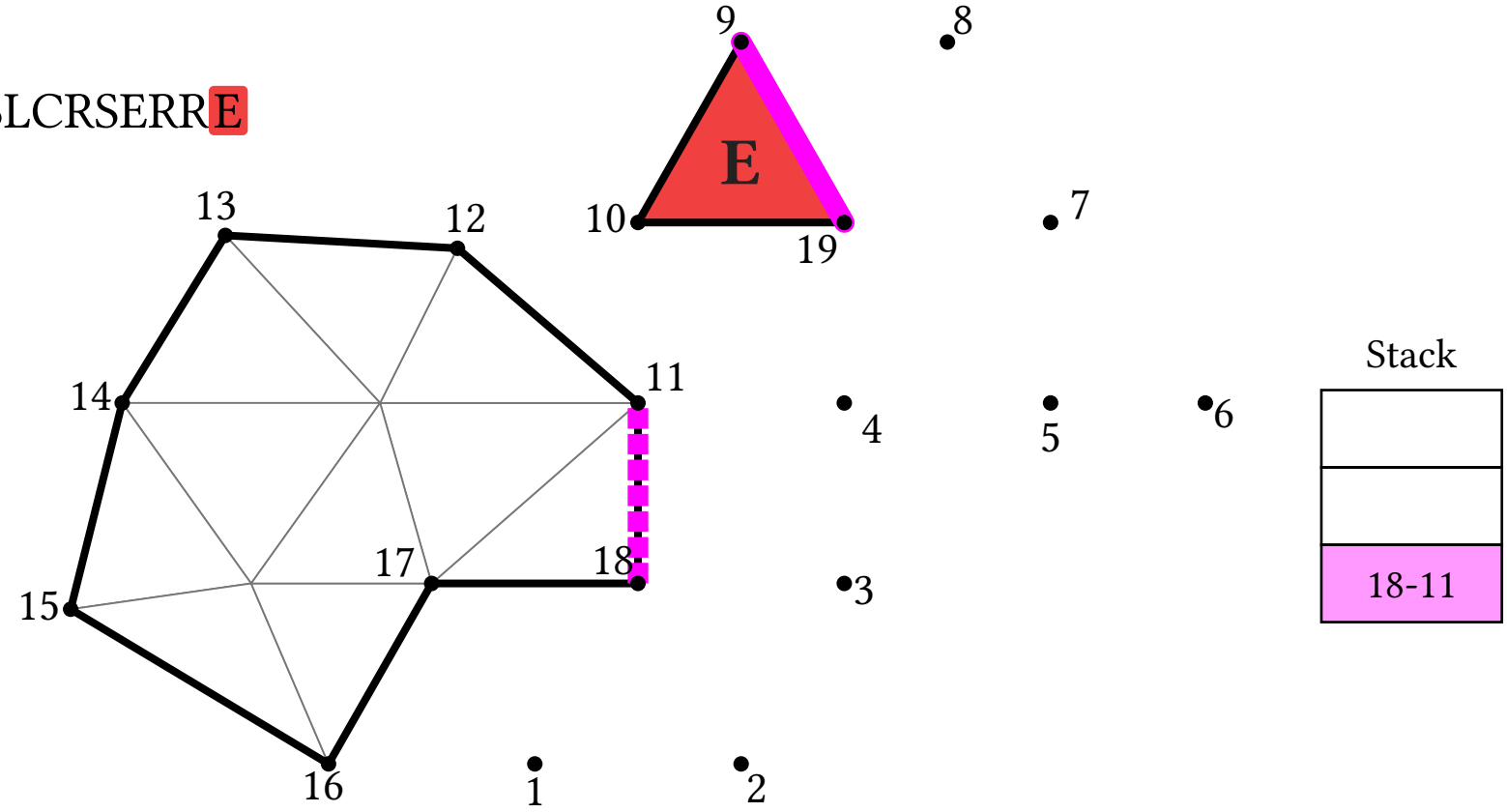


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERREL

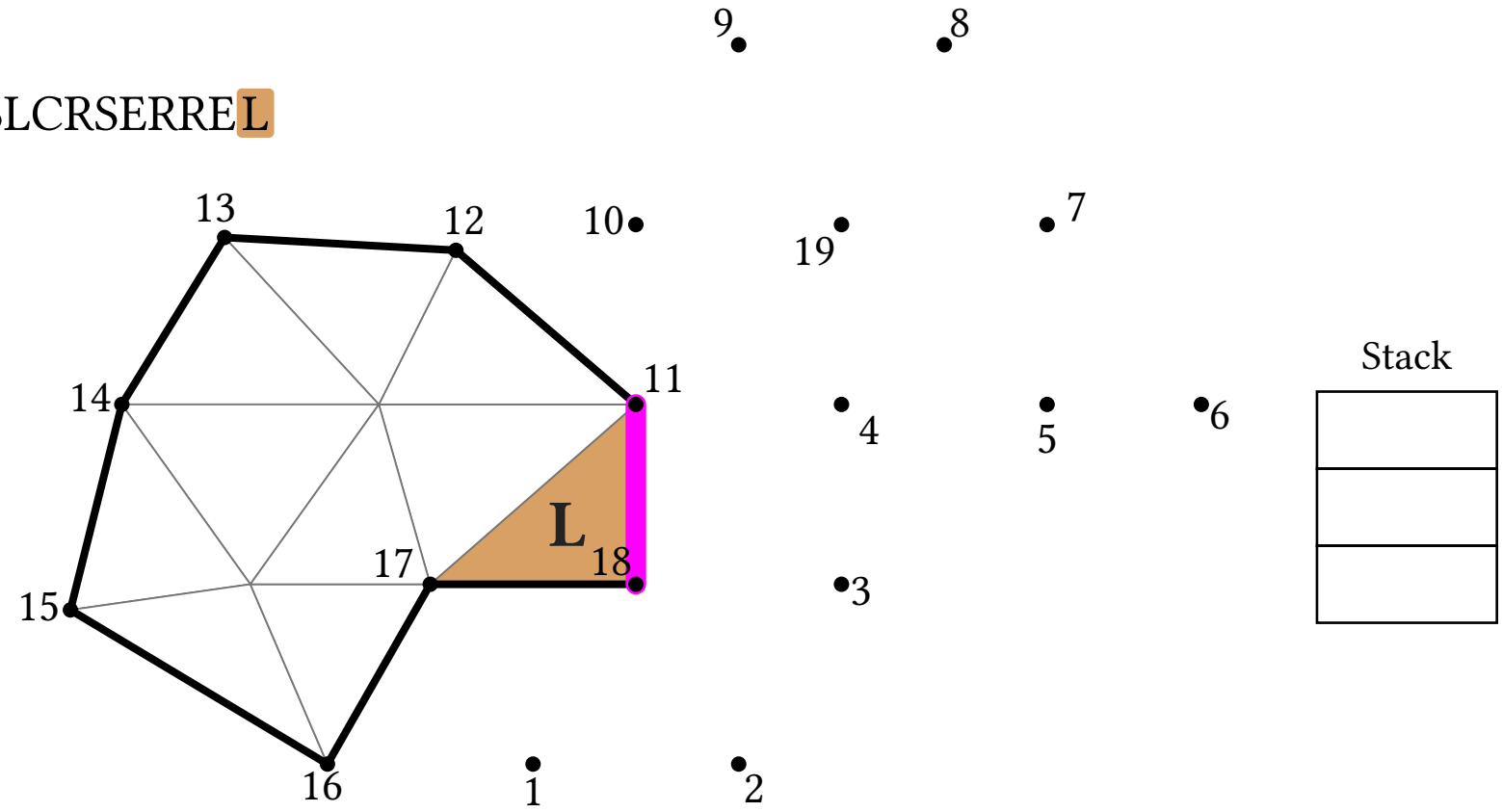


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERREL**C**

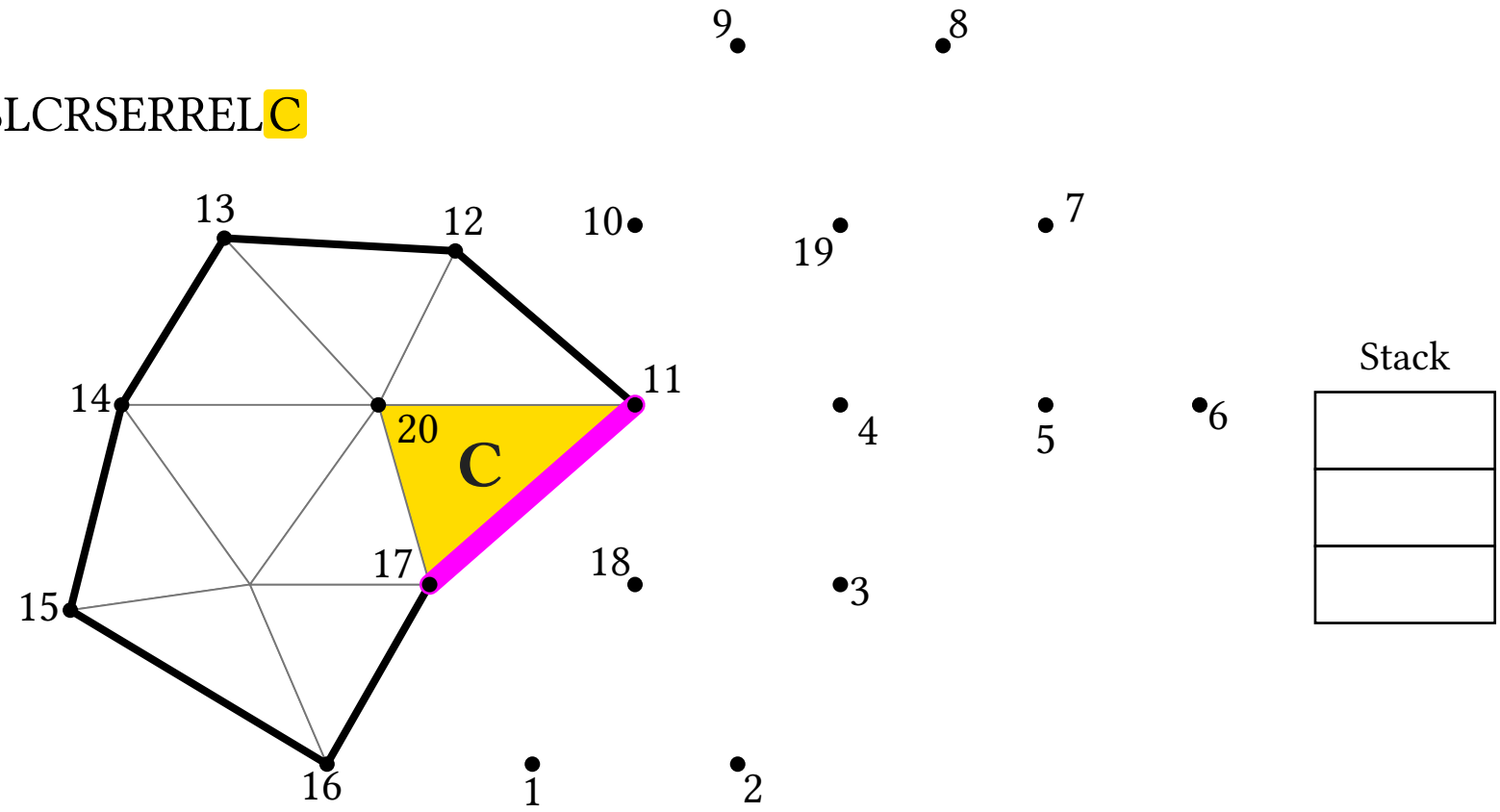


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCR

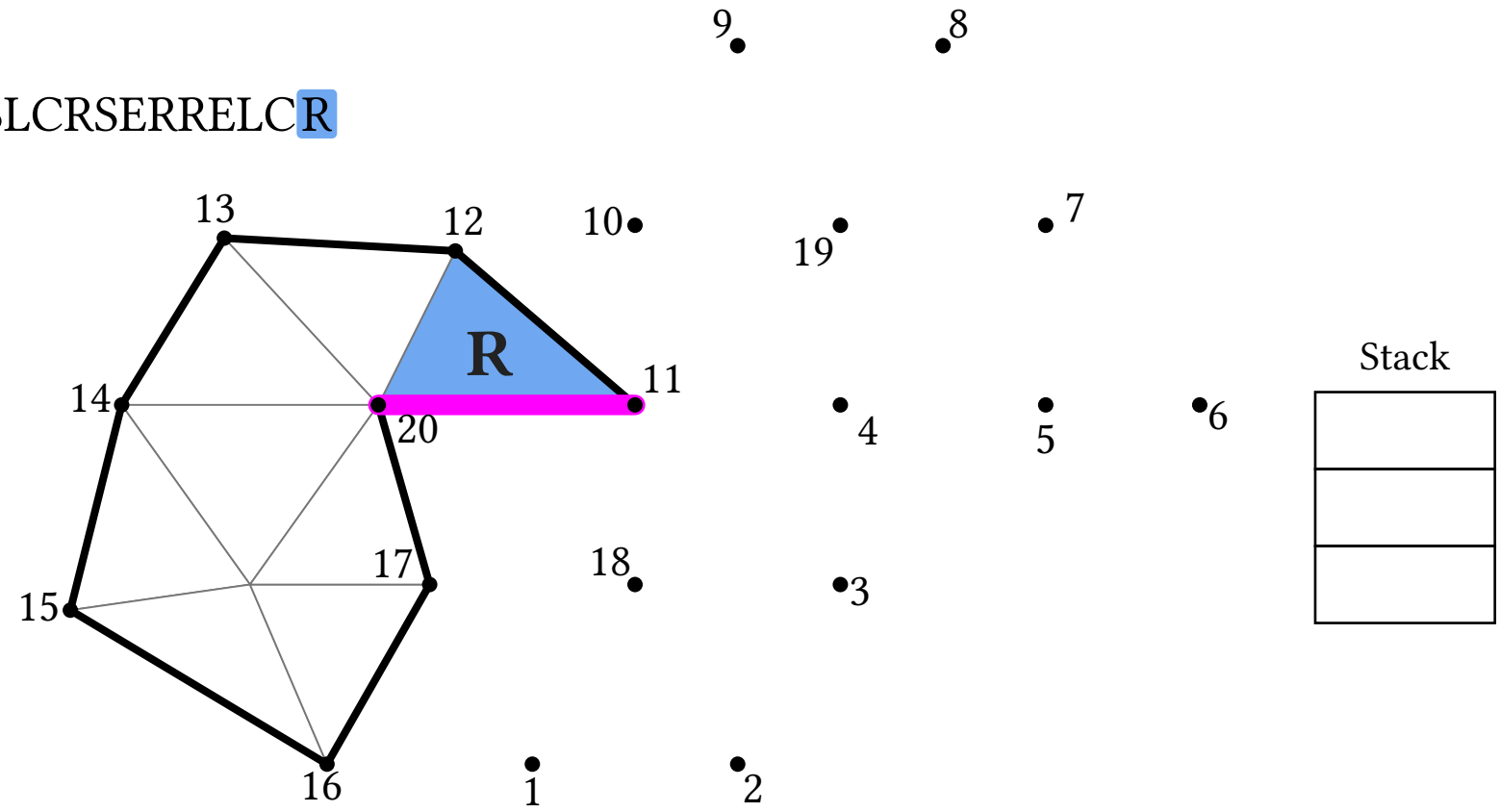


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRR

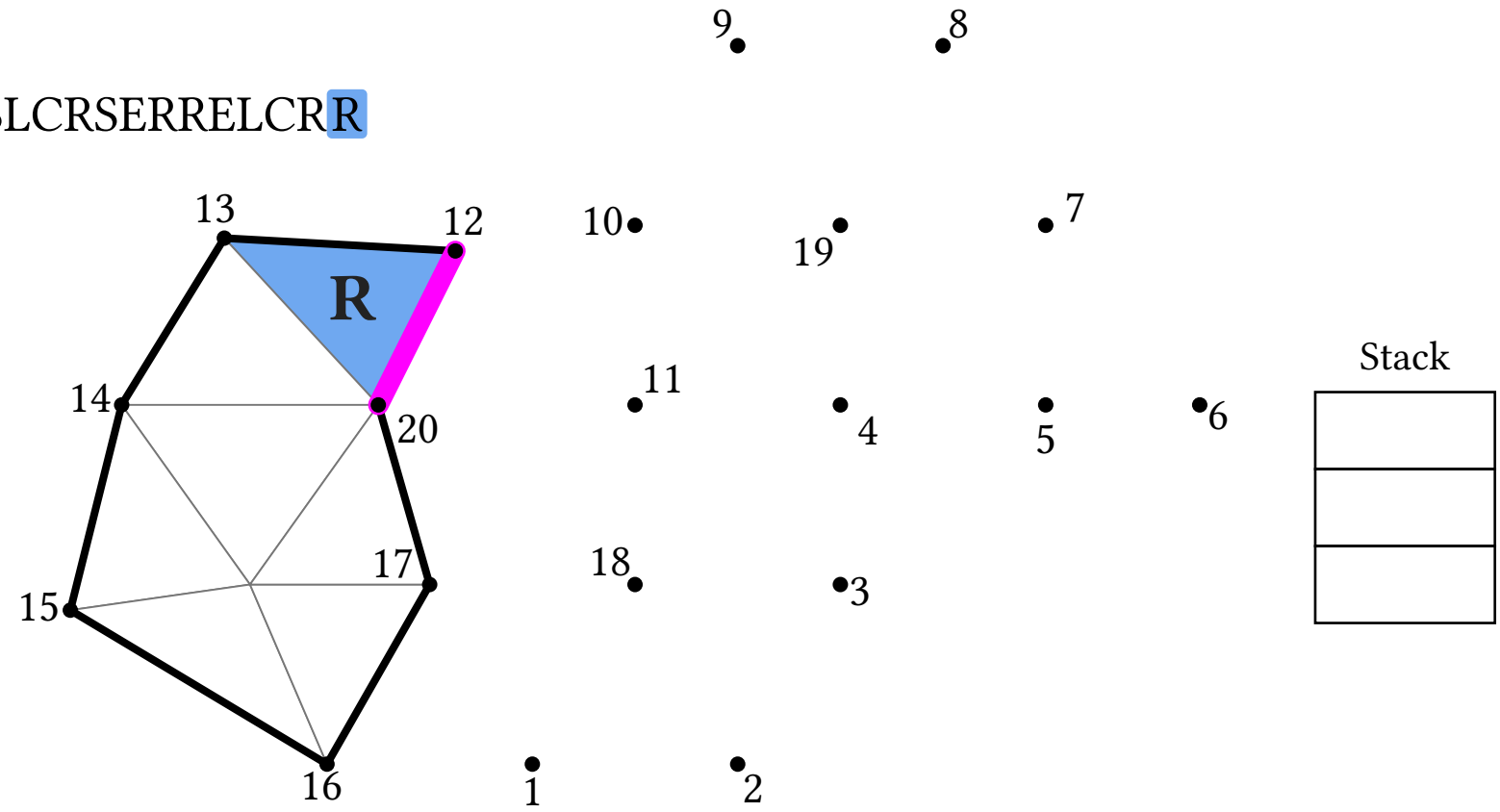


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRR

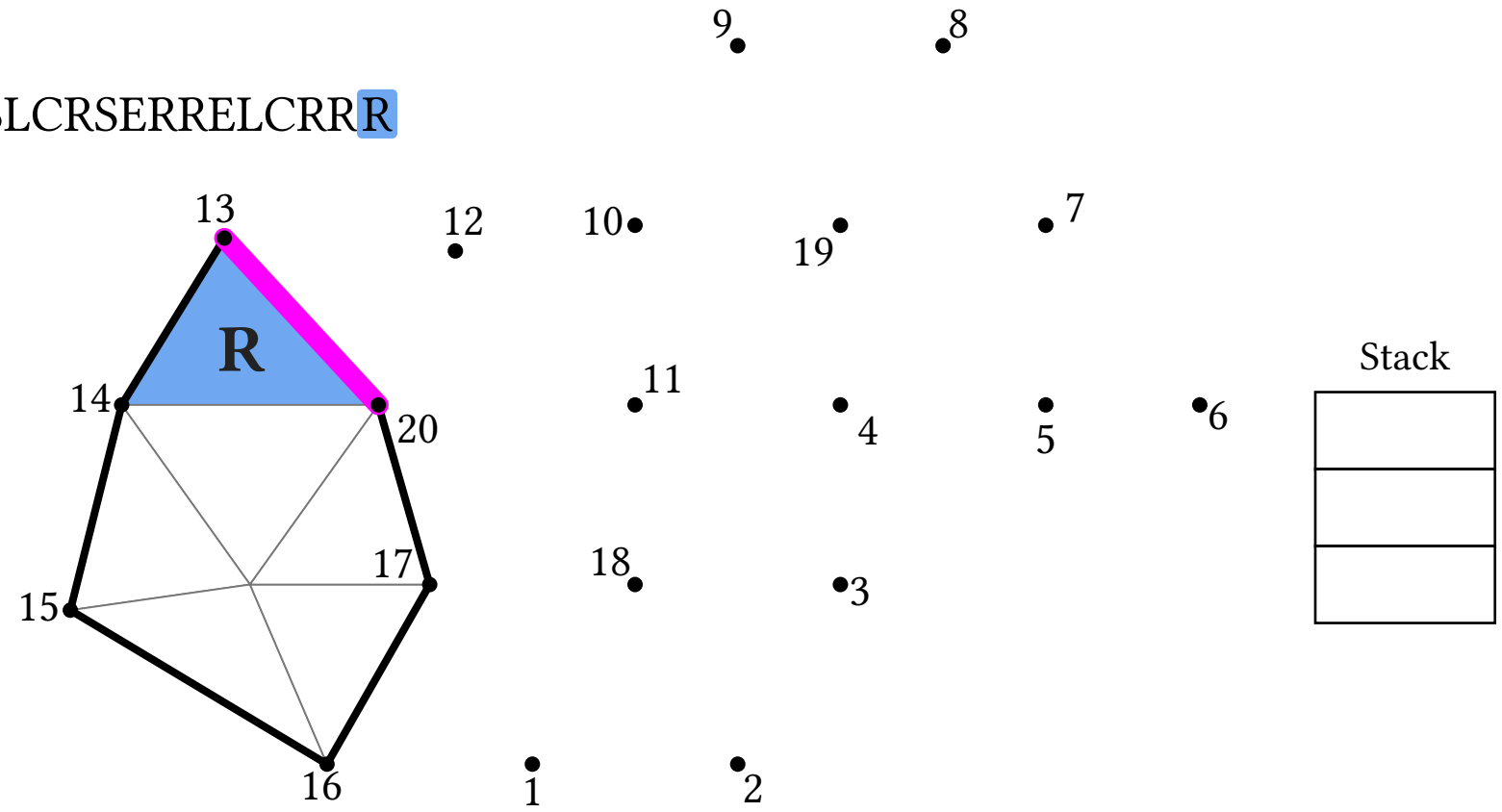


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRC

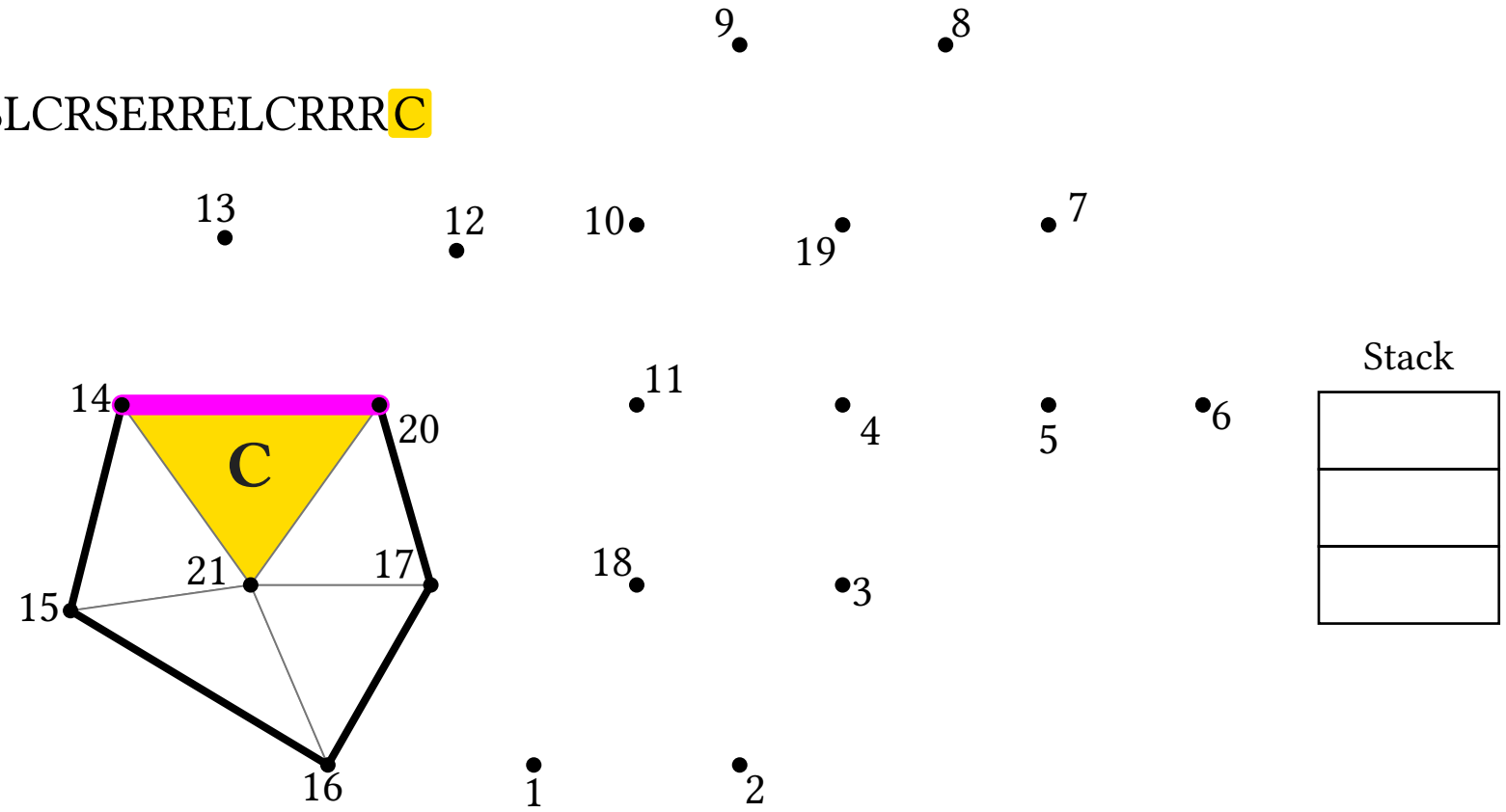


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCR

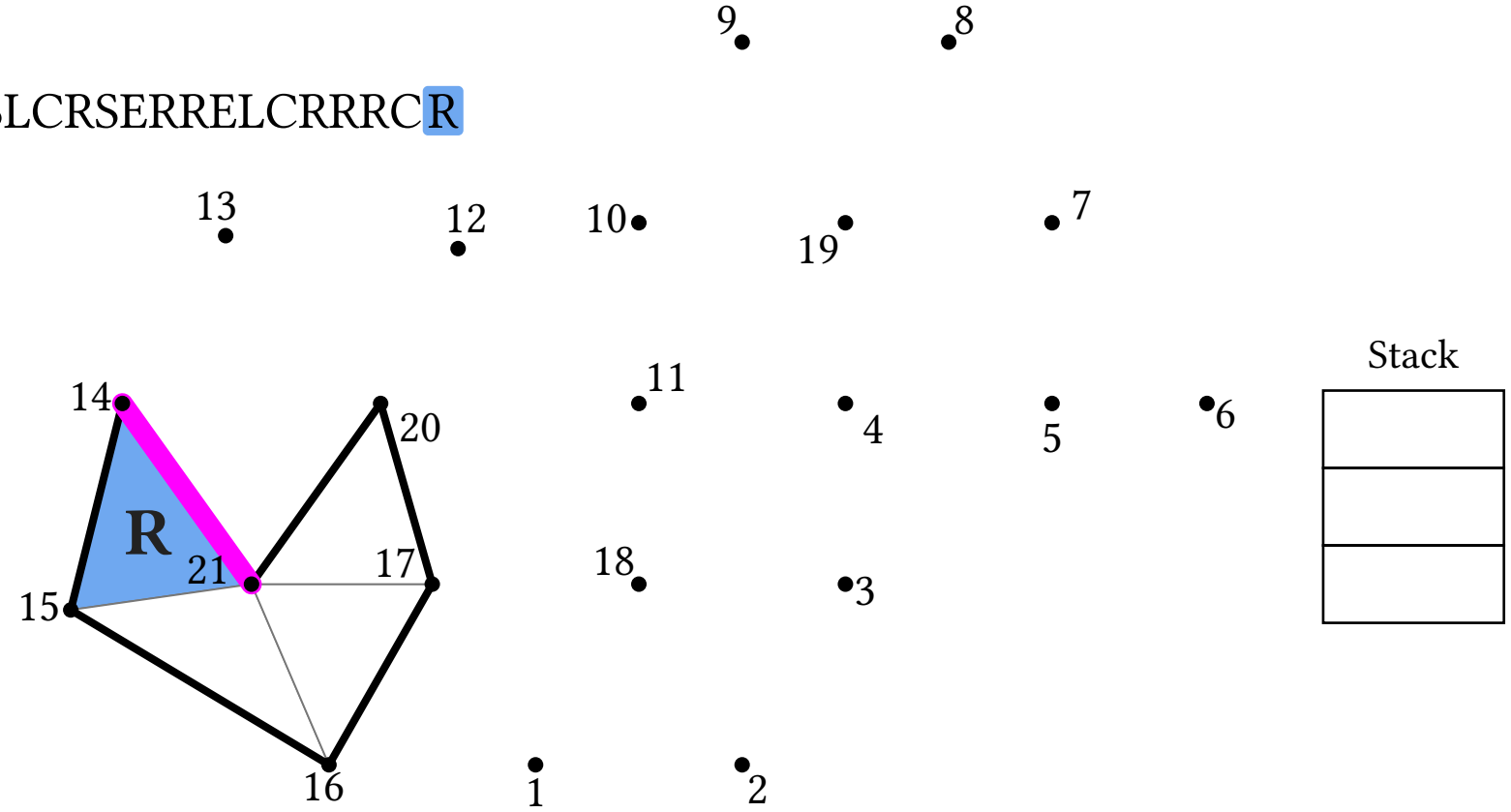


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRR

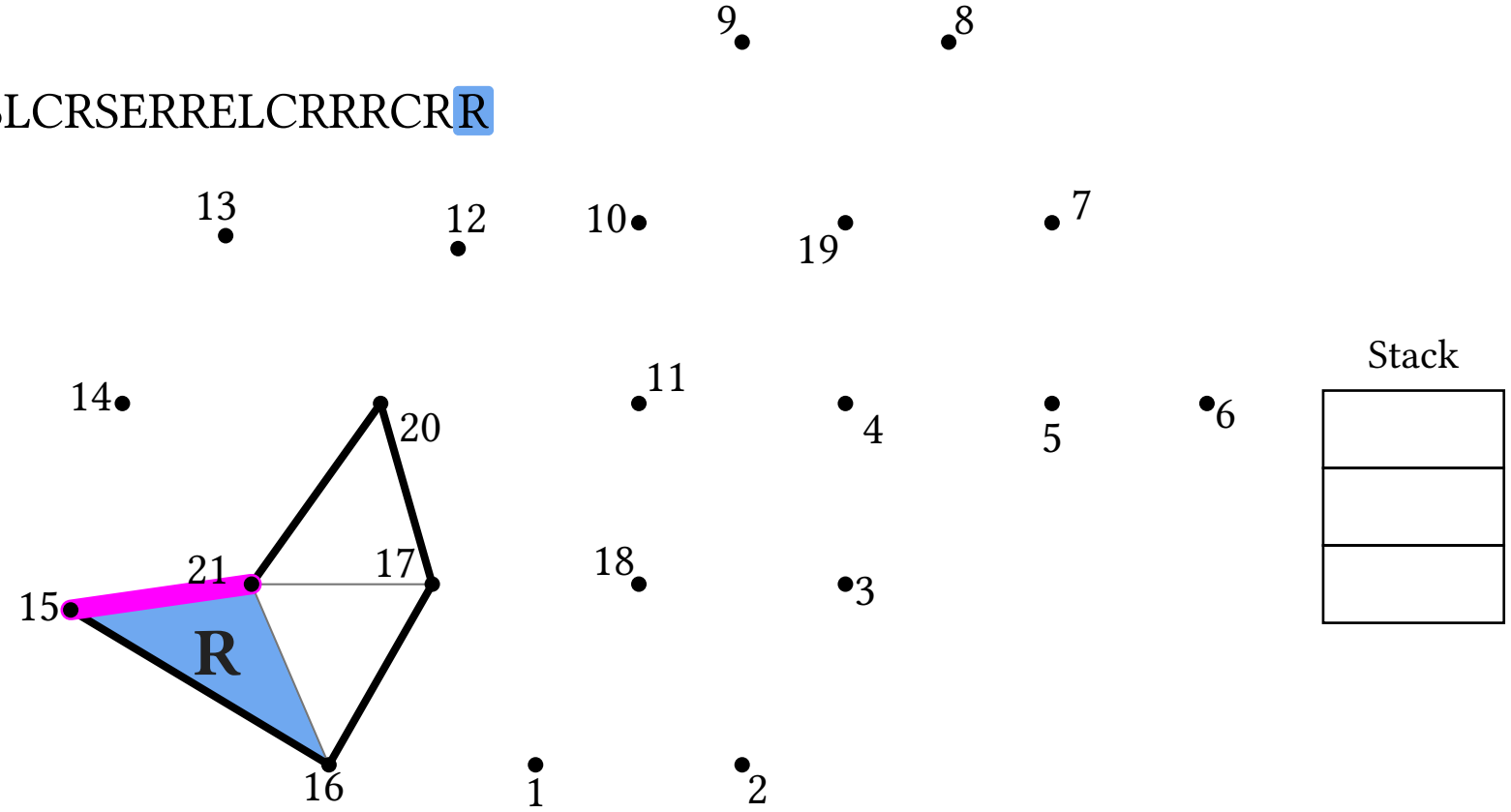


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRR

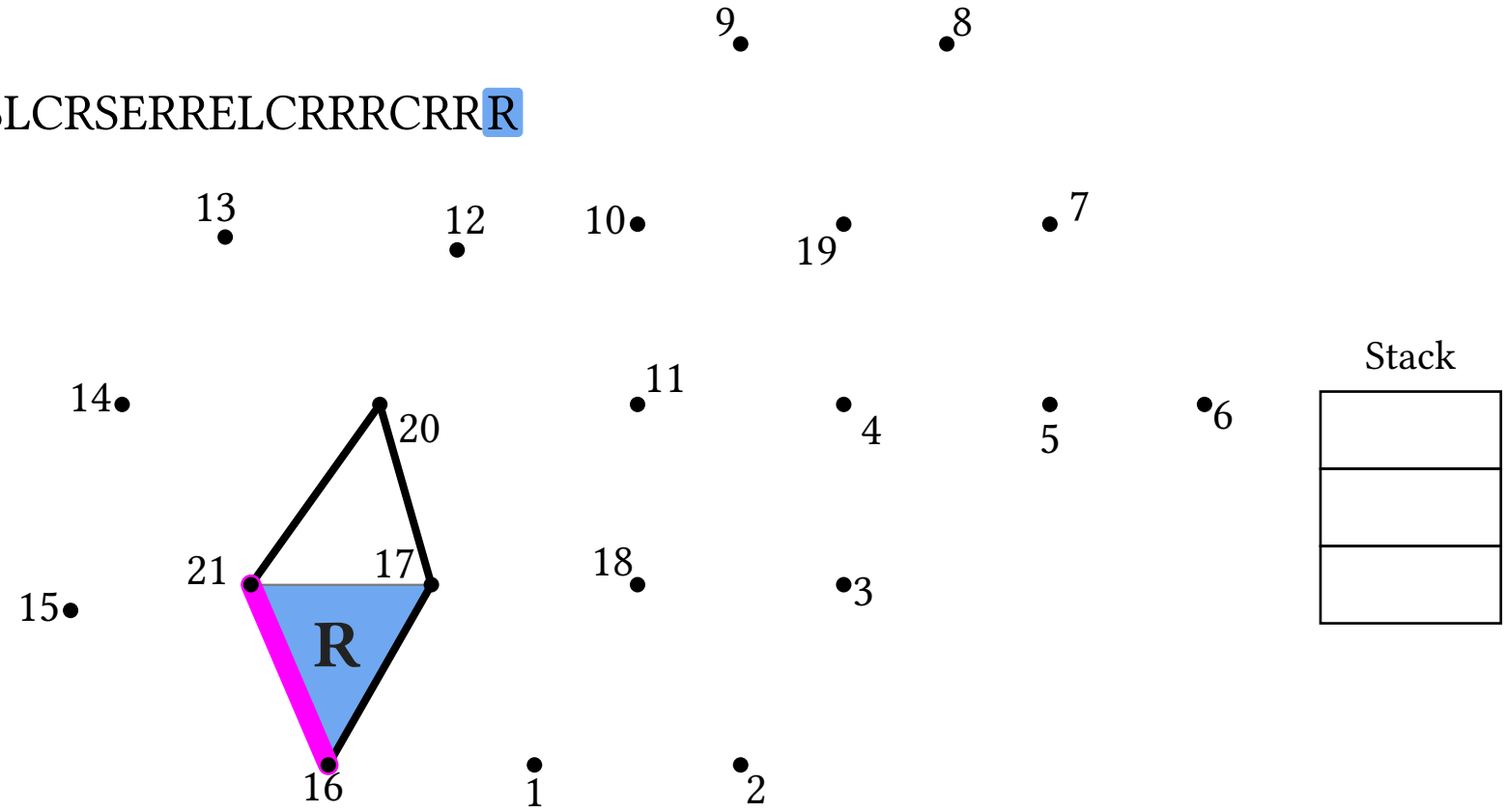


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

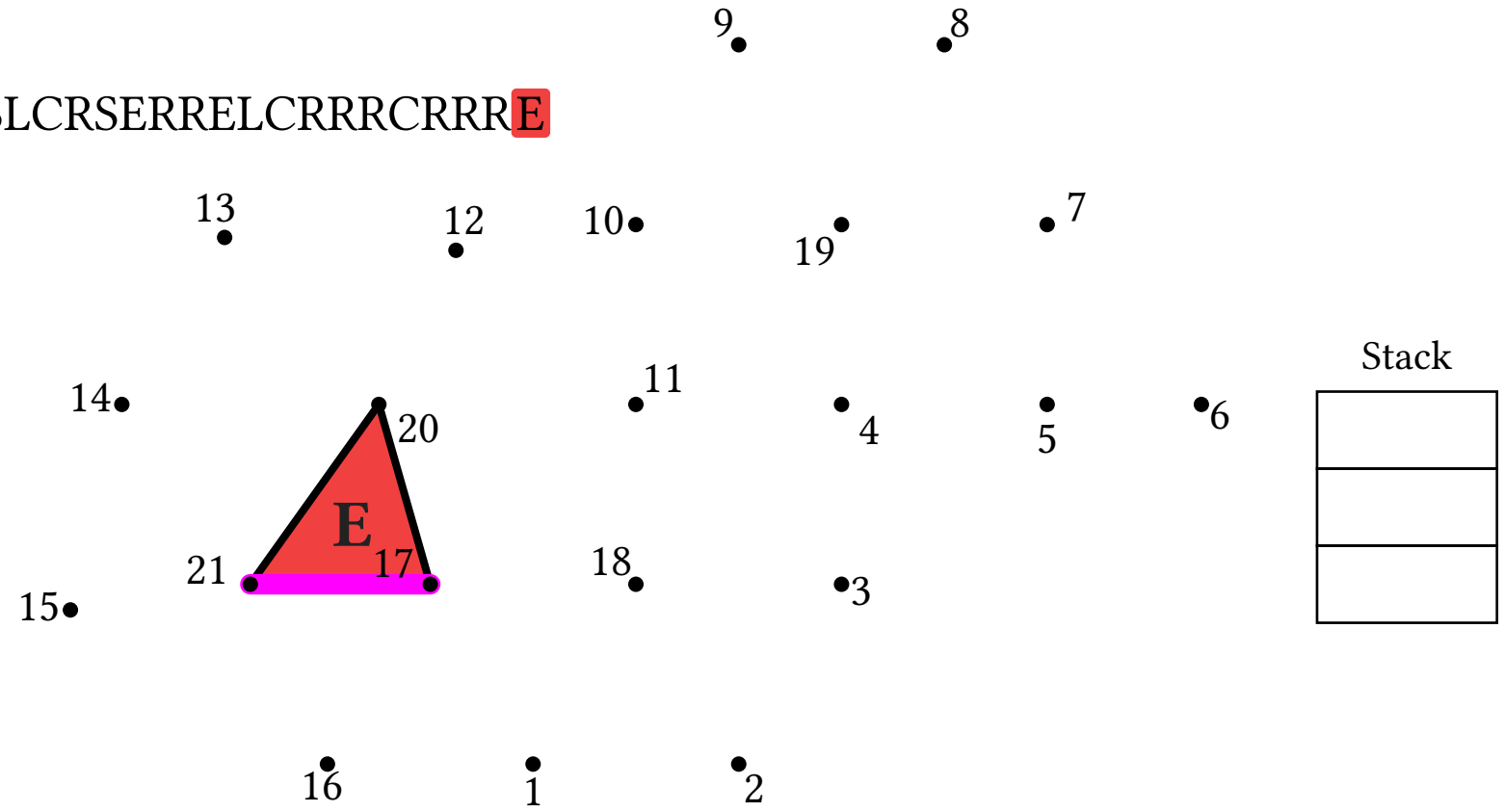


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

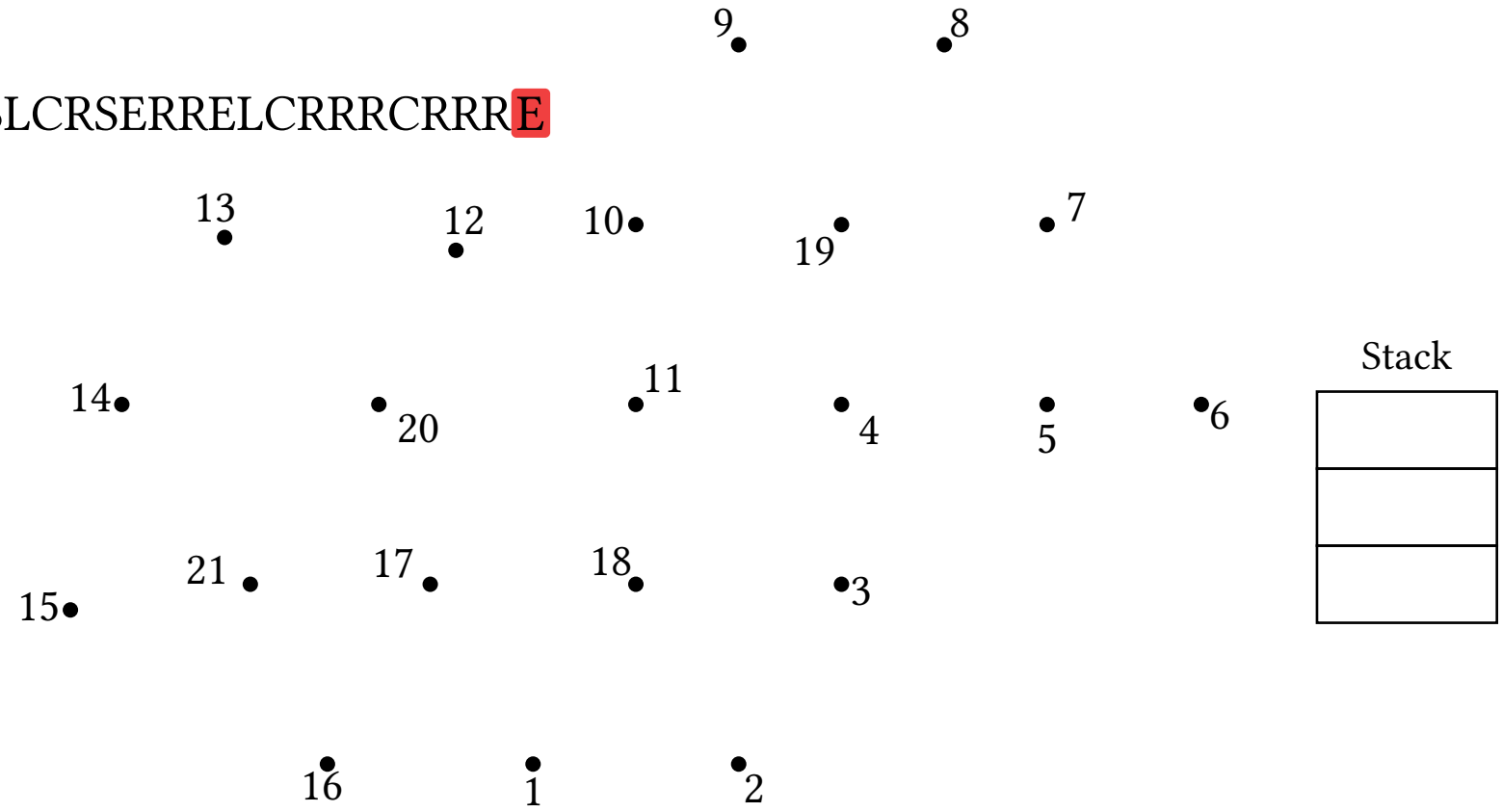


Figure 3: Jednoduchá komprese.

2.3 Vyšší genus?

History =

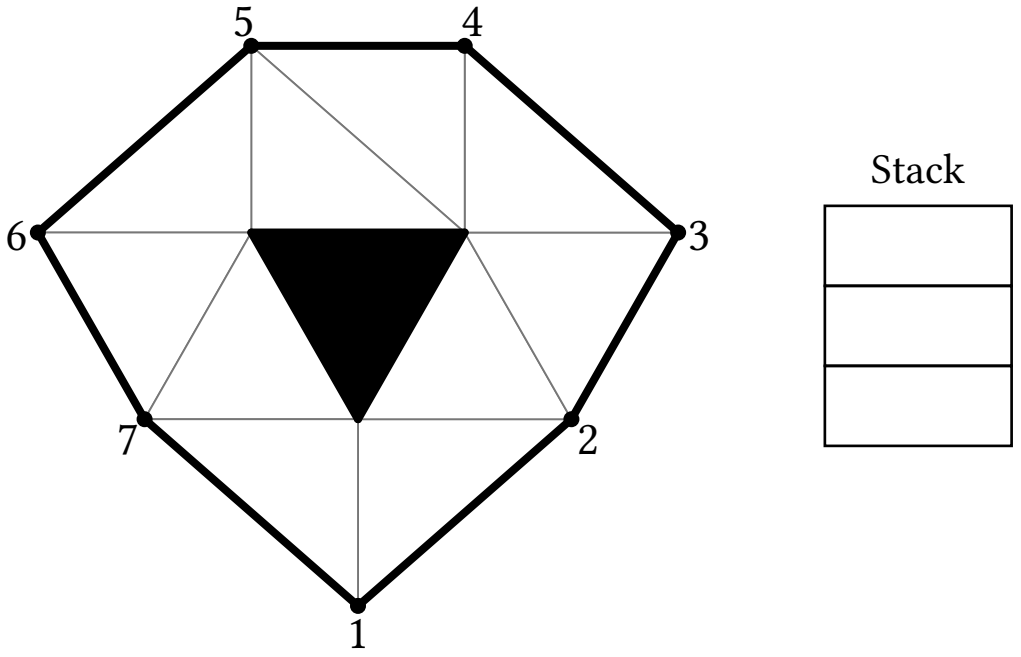


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = S

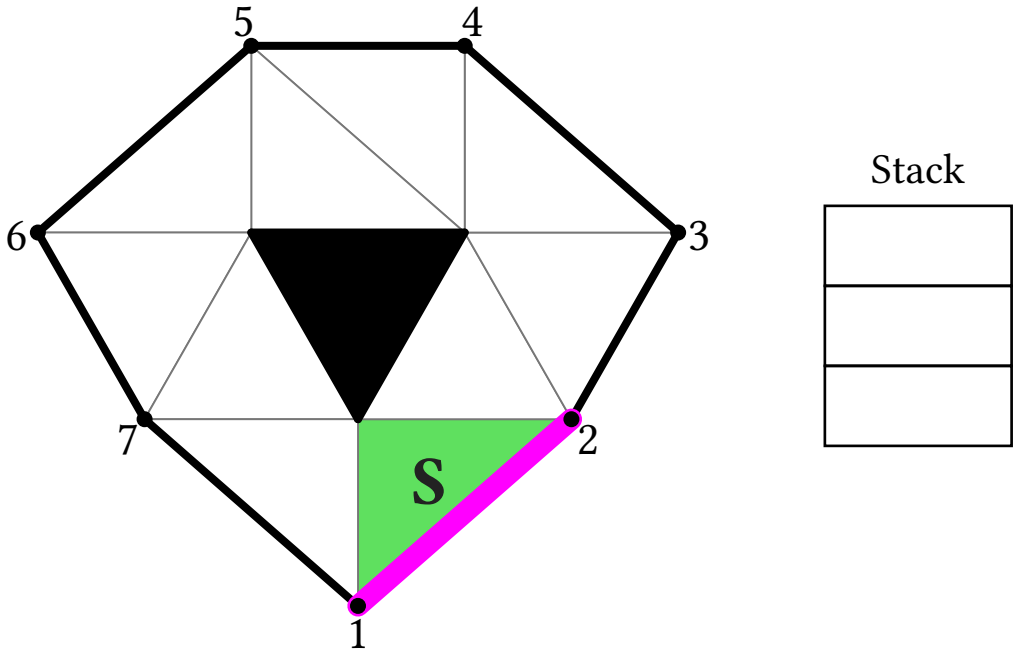


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = S

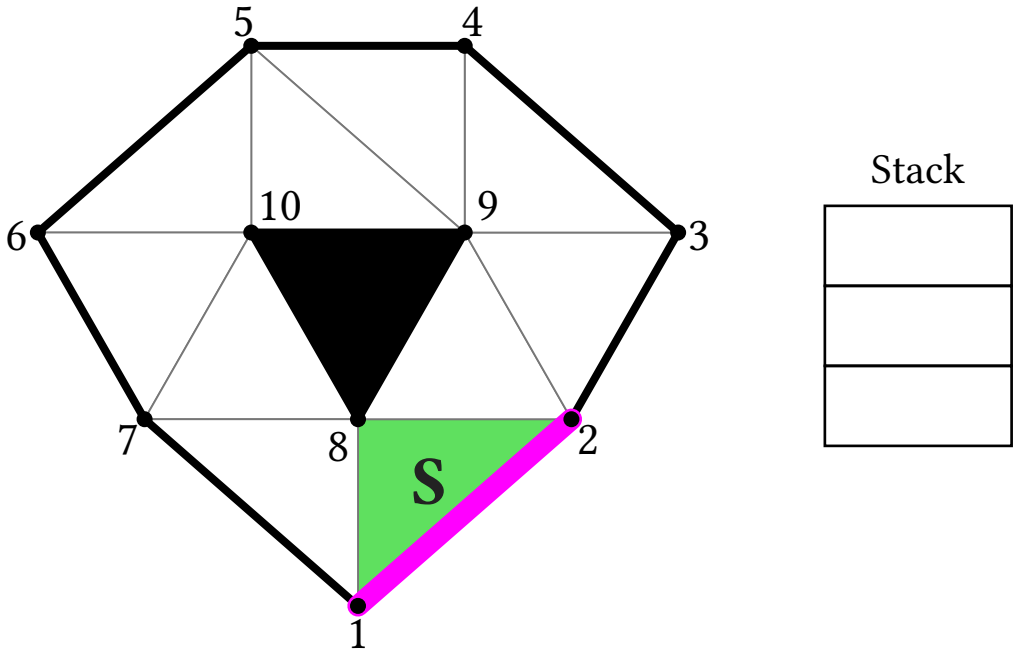


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SL

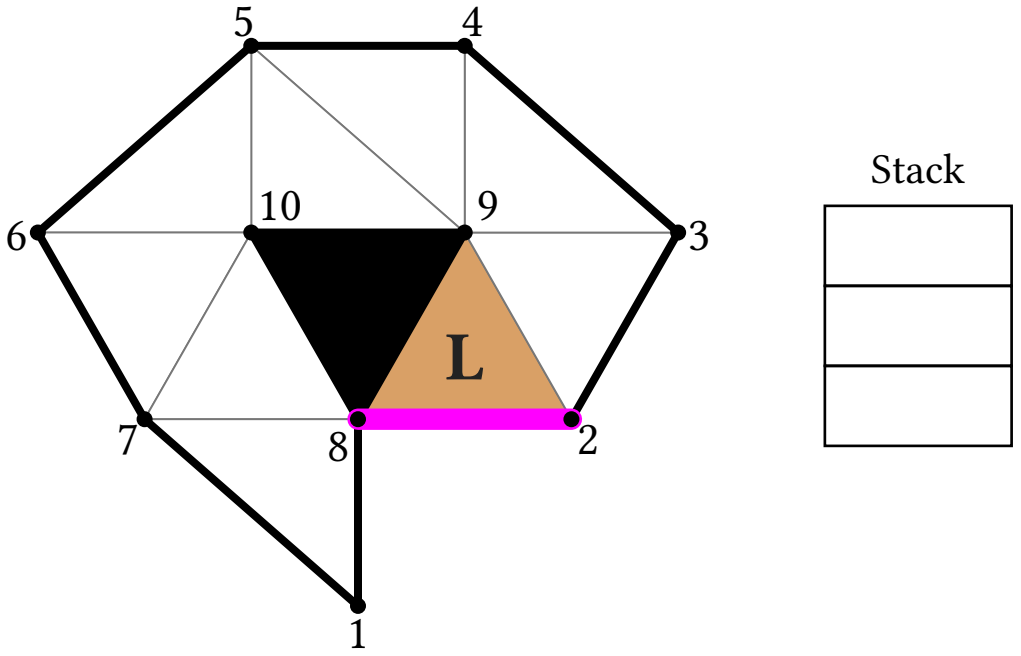


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLR

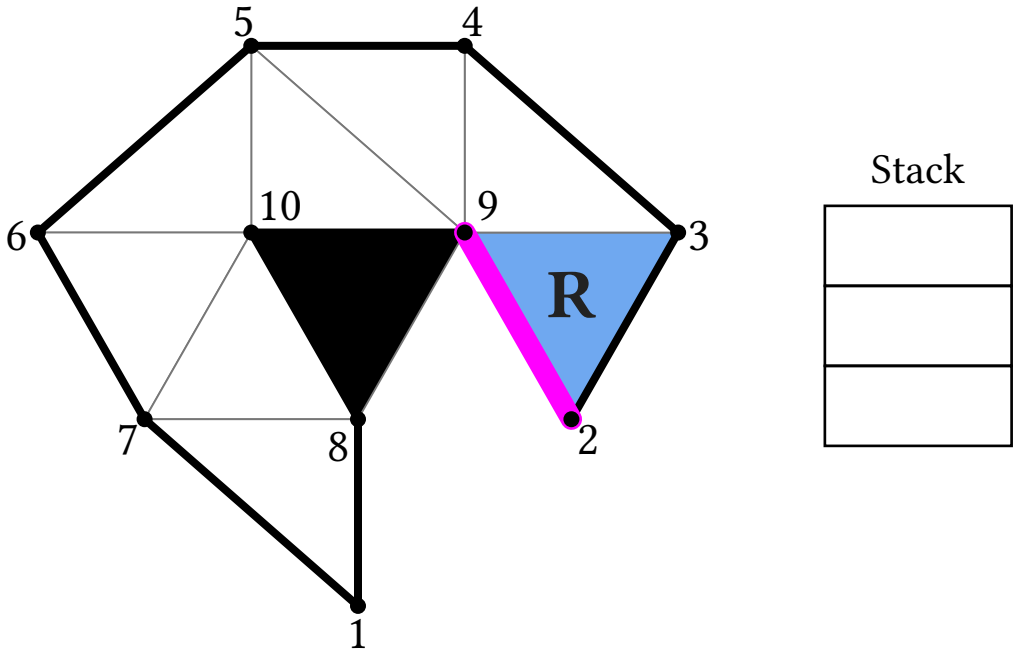


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRR

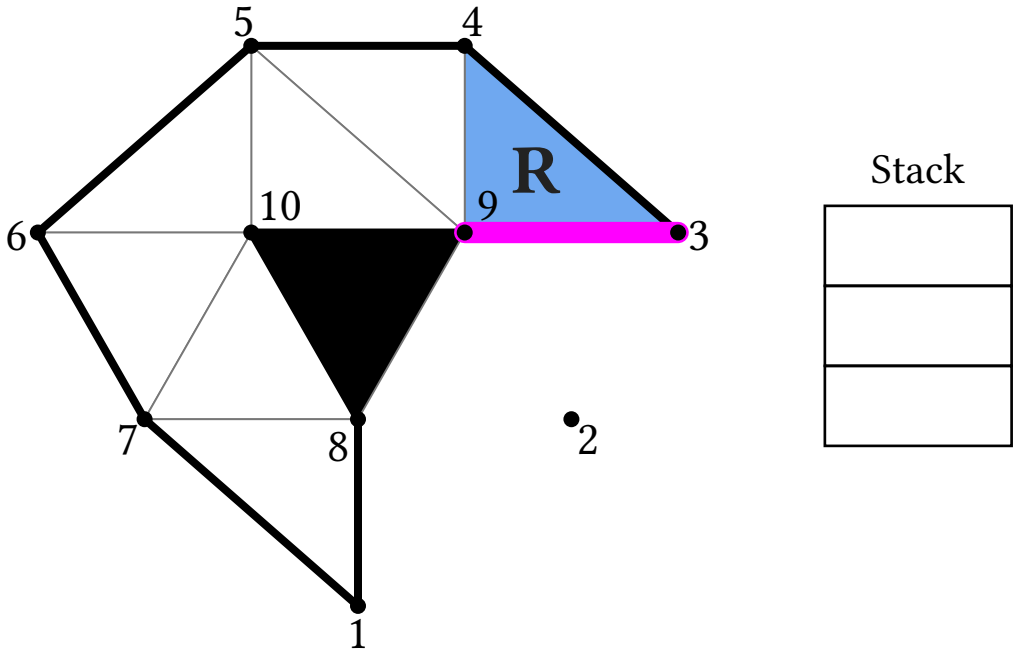


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRR

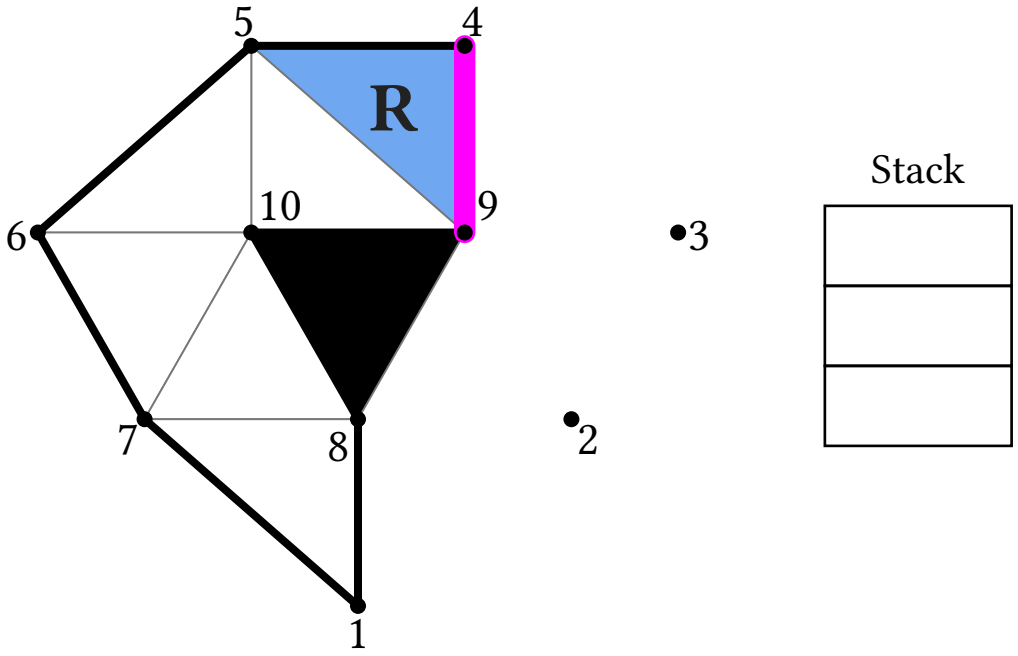


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRL

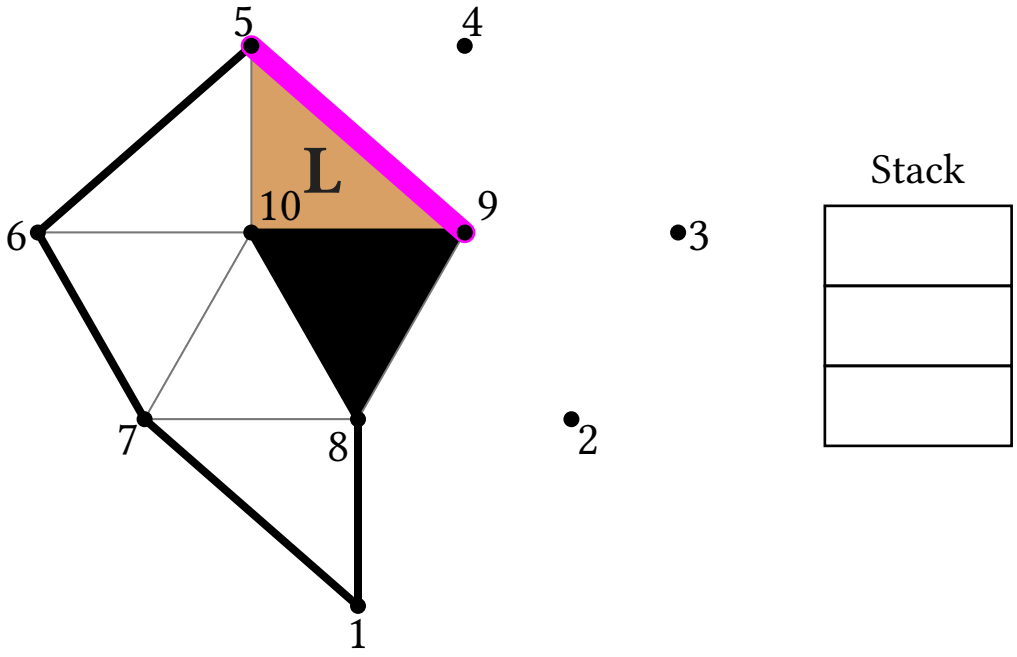


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLR

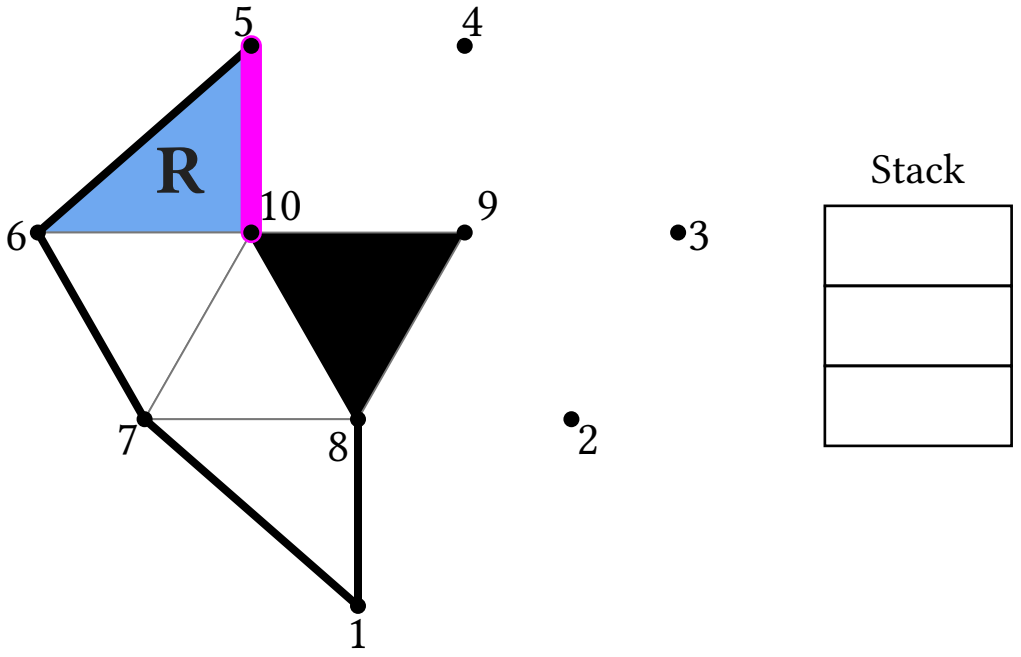


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRR

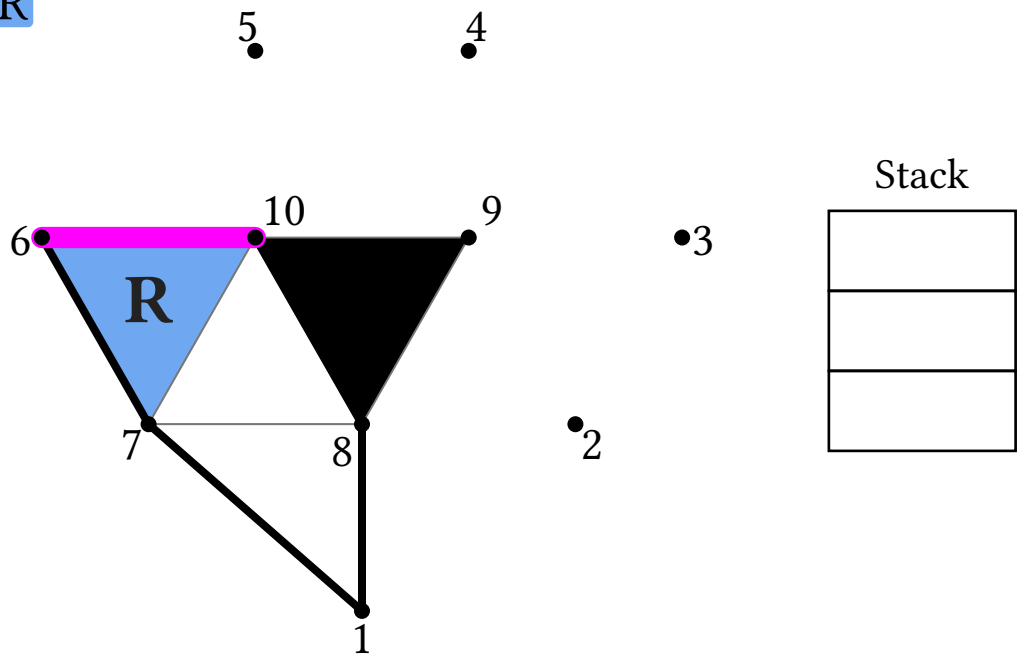


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRRL

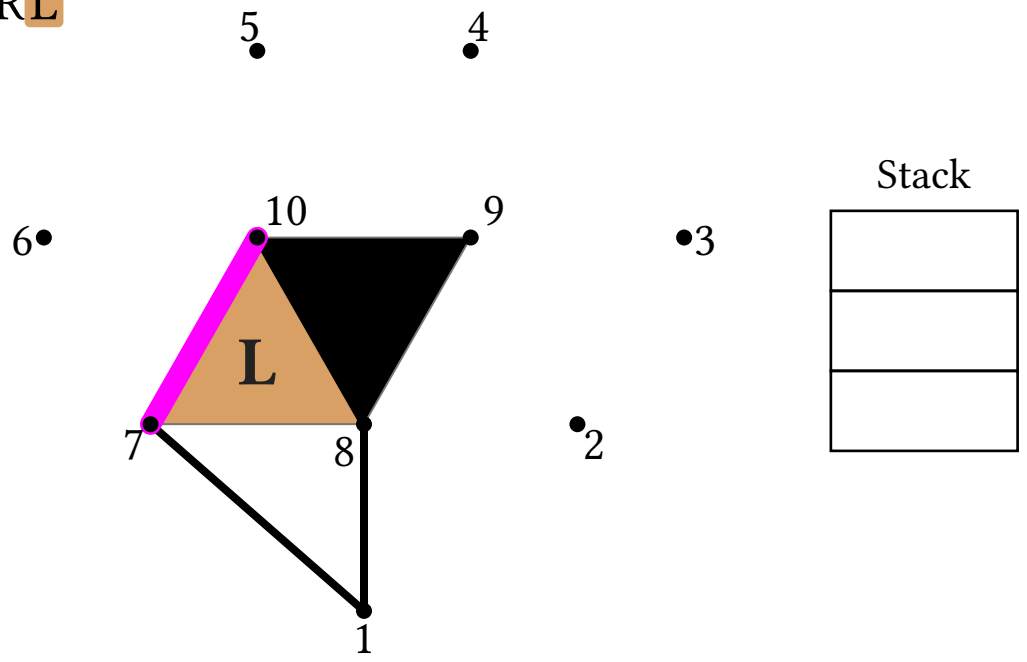


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRRLE

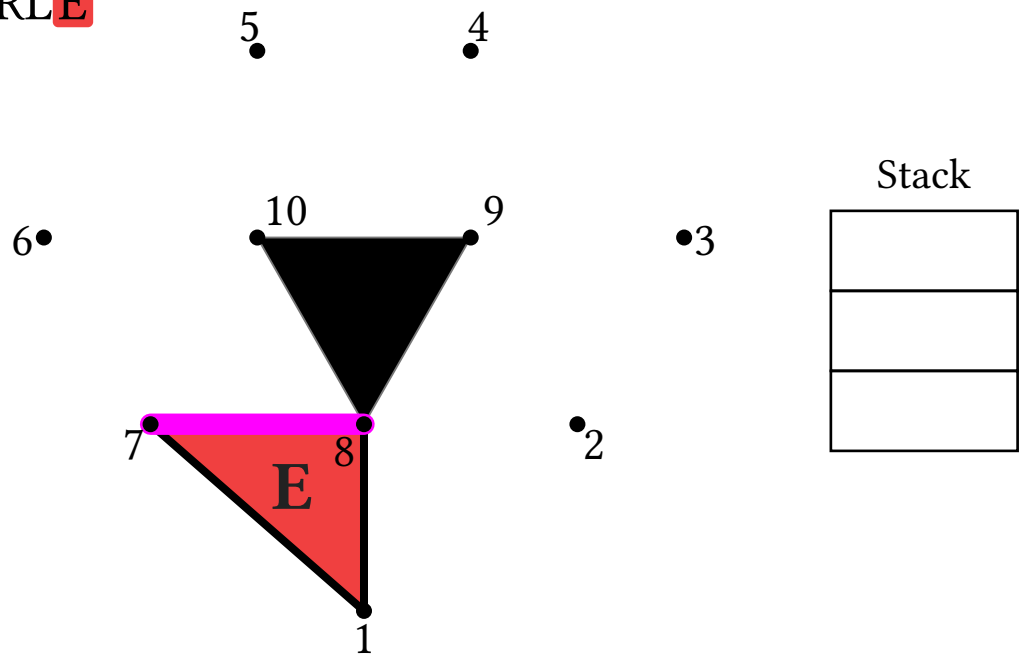


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRRLE

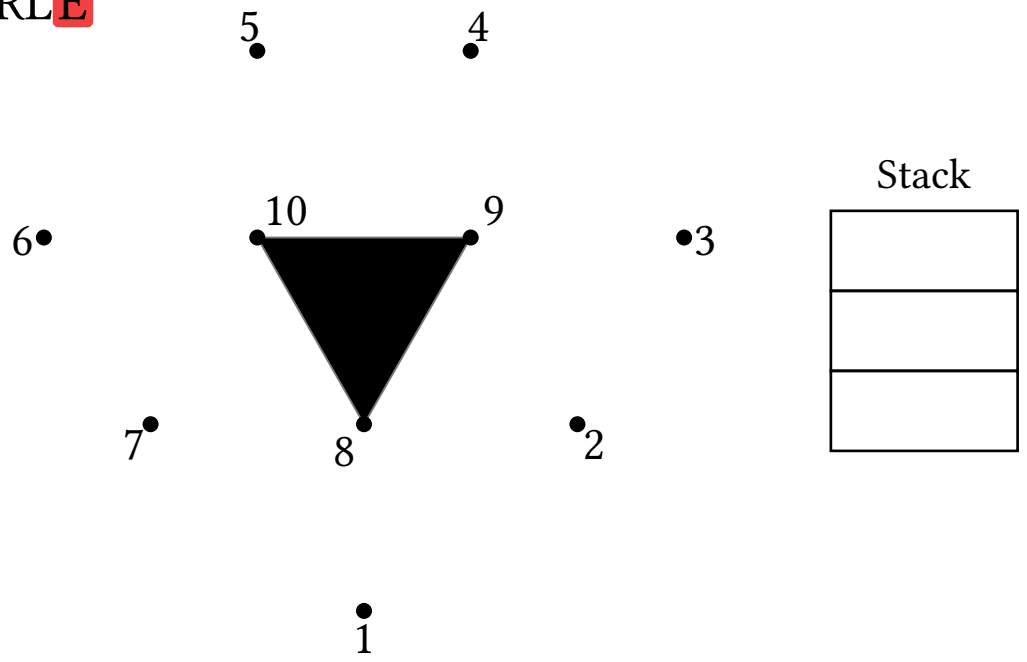


Figure 4: Komprese díry.

3. Dekomprese

3.1 Možnosti

3.2 Pravidla

History = CCRRRSLCRSERRELCRRRCRRRE

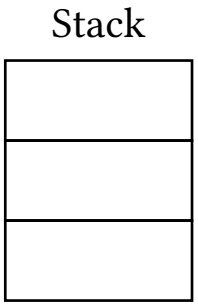


Figure 5: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE

Stack

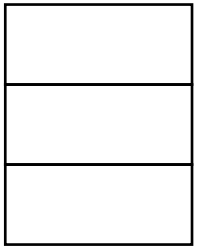
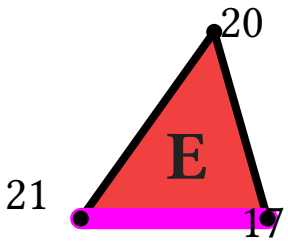


Figure 5: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE



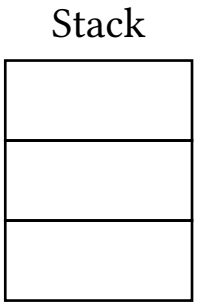
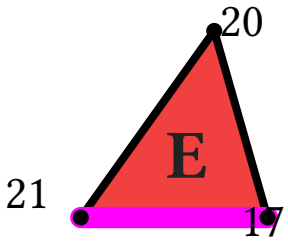


Figure 5: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRR



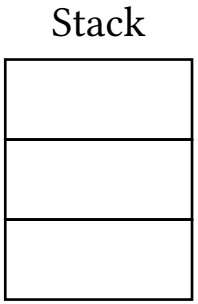


Figure 5: Dekomprese.

4. Zdroje

4.1 Seznam figures

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Figure 3: Jednoduchá komprese. 7
Figure 4: Komprese díry. 8
Figure 5: Dekomprese. 12

4.2 Obrázky (dobový kontext)

1	**Nagano** (2/1998)	Wikipedia, *fair use*
2	**Duda** (4/1998)	Gov.pl, CC BY 3.0 PL, Wikimedia Commons
3	**Windows 98** (5/1998)	SoundCloud (artwork)
4	**MrBeast** (5/1998)	YouTube (avatar kanálu)
5	**Google** (9/1998)	logodix.com
6	**Half-Life** (11/1998)	Valve, valvearchive.com Wikipedia *fair use*
7	**NATO** (3/1999)	nato.int
8	**Futurama** (3/1999)	ČSFD (plakát)
9	**SpongeBob** (5/1999)	craiyon.com
10	**Napster** (6/1999)	Jusore, GPLv3, Wikimedia Commons
11	**Putin** (12/1999)	Kremlin.ru, CC BY 4.0, Wikimedia Commons

4.3 Reference

Originální paper [1] Corner table [2] Významné vylepšení [3] Wrap & Zip [4] Inspirace animací [5]

[1] J. Rossignac, “Edgebreaker: Connectivity Compression for Triangle Meshes,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 5, no. 1, pp. 47–61, Jan. 1999, doi: 10.1109/2945.764870.

[2] J. Rossignac, A. Safonova, and A. Szymczak, “Edgebreaker on a Corner Table: A Simple Technique for Representing and Compressing Triangulated Surfaces,” in *Hierarchical and Geometrical Methods in Scientific Visualization*, G. Farin, B. Hamann, and H. Hagen, Eds., Berlin, Heidelberg: Springer, 2003, pp. 41–50. doi: 10.1007/978-3-662-07443-5_3.

[3] T. Lewiner, H. Lopes, J. Rossignac, and A. W. Vieira, “Efficient Edgebreaker for Surfaces of Arbitrary Topology,” in *Proceedings of SIBGRAPI 2004 (XVII Brazilian Symposium on Computer Graphics and Image Processing)*, Curitiba, Brazil, 2004, pp. 218–225. doi: 10.1109/SIBGRA.2004.1352965.

[4] J. Rossignac and A. Szymczak, “Wrap&Zip: Linear Decoding of Planar Triangle Graphs,” technical report GIT-GVU-99-08, Feb. 1999.

[5] “Edgebreaker: Connectivity Compression for Triangle Meshes.” Accessed: Nov. 03, 2026. [Online]. Available: https://graphics.stanford.edu/courses/cs468-08-winter/slides/sid2.pdf